Analog Electronics

Previous Year Questions

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Analog Electronics (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

Himanshu Agarwal

- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

Anish Saha

- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

Easy Medium Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

Important Links & Resources

Visit our Website
prepfusion.in

Solutions, practice & report
pyq.prepfusion.in

Subscribe: English Channel
youtube.com/@PrepFusion_GATE

Subscribe: Hindi Channel
youtube.com/@prepfusion-hindi

Join our Telegram group
t.me/All_About_Learning

Follow us on LinkedIn
linkedin.com/company/prepfusionee

Subscribe: Himanshu Agarwal
youtube.com/@HimanshuAgarwal_

Subscribe: Anish Saha
youtube.com/@AnishSaha_

Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

This book exists because of the people below, who built, reviewed and typeset its content.

Design & Layout Architect — Nawras Ahamed.

Technical Reviewers — Ayush Prakash Singh, Abhishek Bhattacharjee, Shujaut Yaseen, Yuva Kishore Gottapu, Manepalli Pavan Vignesh Kumar, Palak Agarwal, Devmalya Ghatak, Kasi Viswanath A, Suresh Ravanam, Aabhas Anuraag Suman and Piyush Raj.

Content Developers — A S Srimithra, Chinthada Lakshmi Prasad, Gantasala Ravi Teja, Guduru Vinayak Varma, Harshaditya Punera, Katikala Sarath, Pranitha Musunuru, Safdar Nazim, Sai Mahalakshmi Janjanam, Shaila Daniels, Sharvari Patwardhan, Sumedha Ghosh, Vinay K Gowda, Aayushi Ghosh, Chevuri Siva Naga Sai Abhi Chandar, Govind Sharma, Kasina Sai Surya Vamsi, Machanpally Karthik Kumar, Seeku Vamsi Krishna and Suprio Dutta.

Contents

Sl.	Topic	Pages
1	Diode Circuits and applications	1
2	MOSFETs	8
3	Operational Amplifier	13
4	BJTs	42
5	Miscellaneous Topics	50

PrepFusion

UNIT

I

Diode Circuits and applications

Topics

1. Diode Models with open and short circuit tests
2. Operating Point and Small Signal Analysis of a Diode
3. Zener Diode as a Voltage Regulator
4. Series and Shunt Clipper Circuits
5. Two-Level Clipper Circuits
6. Clamper Circuits and their examples
7. Peak Detector and Average Diode Current
8. Voltage Multiplier Circuits
9. Half Wave Rectifier and Centre Tapped Full Wave Rectifier
10. Full Wave Bridge Rectifier and Capacitive Filters

Unit Snapshot*

Stream: IN	**Questions:** 27	**MCQ:** 18	**MSQ:** 0	**NAT:** 9
1M: 18	**2M:** 9	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–5M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT 2026 1M

In the circuit below the maximum value of v_{out} is _____ V. *(rounded off to the nearest integer)*

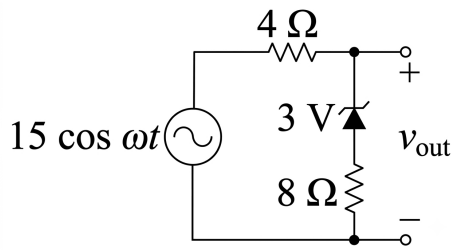


Fig. Q1.26.1

Q1.23.1 MCQ 2023 1M

A light emitting diode (LED) emits light when it is biased. A photodiode provides maximum sensitivity to light when it is biased.

A. forward, forward
B. forward, reverse
C. reverse, reverse
D. reverse, forward

Q1.23.2 NAT 2023 1M

The diode in the circuit is ideal. The current source is $i_s(t) = \pi \sin(3000\pi t)$ mA. The magnitude of the average current flowing through the resistor R is _____ mA *(rounded off to two decimal places).*

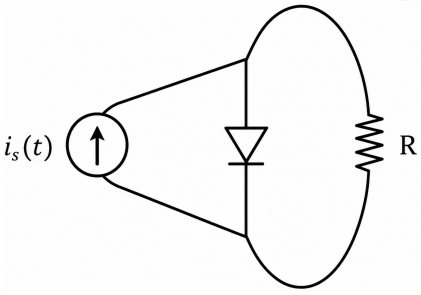


Fig. Q1.23.2

Q1.22.1 NAT 2022 1M

A Zener diode is used as a 4 V voltage regulator in the circuit shown. Given that the diode requires a minimum current of 4 mA for voltage regulation, the maximum current (in milliamperes) permitted to flow through the load R_L is _____ *(round off to one decimal place).*

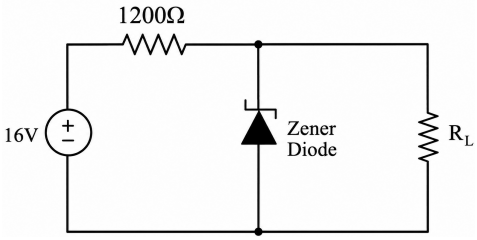


Fig. Q1.22.1

Q1.22.2 MCQ 2022 2M

A signal $V_{in}(t)$ shown is applied from $t = 0$ ms to $t = 6$ ms to the circuit shown. Given the initial voltage across the capacitor is 0.3 V, and that the diode is ideal, the open circuit voltage $V_{out}(t)$ at $t = 5$ ms is _____ .

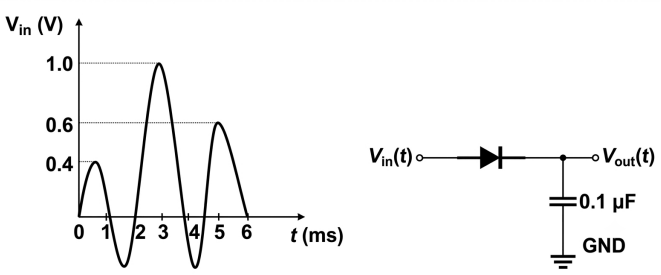


Fig. Q1.22.2

A. 0.3 V **B.** 0.6 V **C.** 0.7 V **D.** 1.0 V

Q1.22.3 NAT 2022 2M

In the circuit shown, the capacitance $C_0 = 10$ μF and inductance $L_0 = 1$ mH and the diode is ideal. The capacitor is initially charged to 10 V and the current in the inductor is initially zero. If the switch is closed at $t = 0$ s, the voltage $V_C(t)$ (in volts) across the capacitor at $t = 0.5$s is _____ *(round off to one decimal place).*

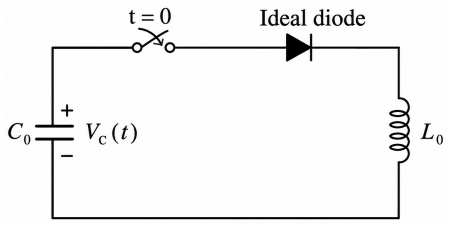


Fig. Q1.22.3

Q1.20.1 MCQ 2020 1M

If the diodes in the circuit shown are ideal and the breakdown voltage V_Z of the Zener diode is 5V, the power dissipated in the 100 Ω resistor (**in watts**) is _____.

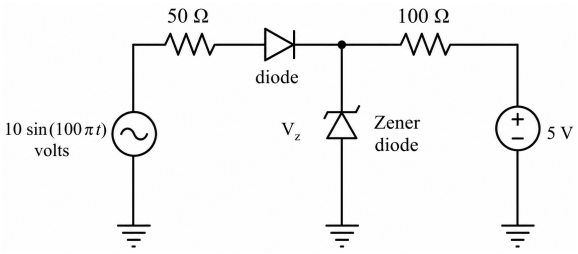


Fig. Q1.20.1

A. 0 **B.** 1 **C.** $\frac{25}{100}$ **D.** $\frac{225}{100}$

Q1.20.2 NAT 2020 2M

Assume the diodes in the circuit shown are ideal.

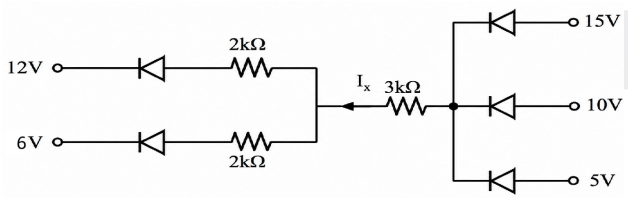


Fig. Q1.20.2

The current I_x flowing through the 3 kΩ resistor (in mA, rounded off to one decimal place) is _____ .

Q1.19.1 MCQ 2019 1M

In the figures (a) and (b) shown below, the transformers are identical and ideal, except that the transformer in figure (b) is centre-tapped. Assuming ideal diodes, the ratio of the root-mean-square (RMS) voltage across the resistor R in figure (a) to that in figure (b) is

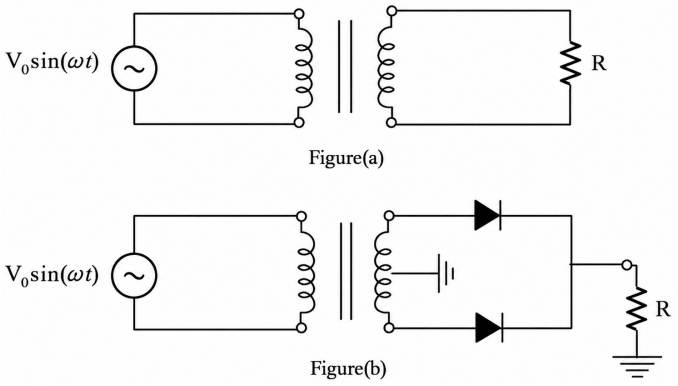


Fig. Q1.19.1

A. $\sqrt{2}:1$ **B.** $2:1$ **C.** $2\sqrt{2}:1$ **D.** $4:1$

Q1.18.1 NAT 2018 1M

The diodes given in the circuit are ideal. At $t = 60$ ms, V_{pq} (in Volts) is _____ .

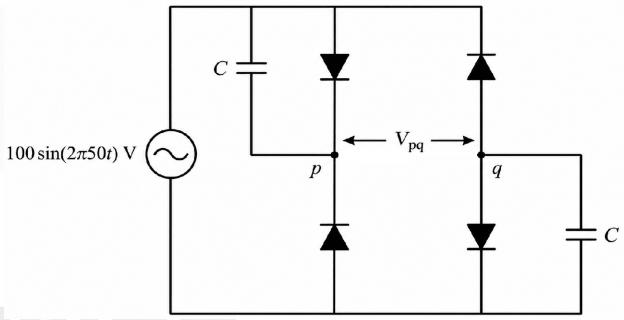


Fig. Q1.18.1

Q1.17.1 MCQ 2017 1M

The silicon diode, shown in the figure, has a barrier potential of 0.7 V. There will be no forward current flow through the diode. If V_{dc}, in volts, is greater than

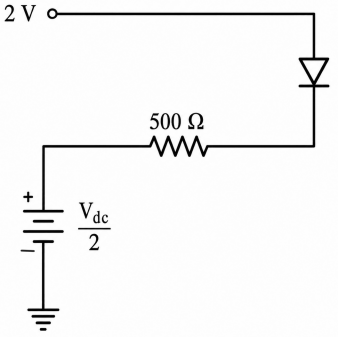


Fig. Q1.17.1

A. 0.7 **B.** 1.3 **C.** 1.8 **D.** 2.6

Q1.16.1 NAT 2016 1M

The diode D used in the circuit below is ideal. The voltage drop V_{ab} across the 1 kΩ resistor in volt is _____ .

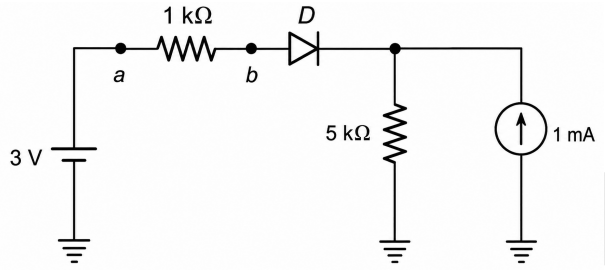


Fig. Q1.16.1

Q1.15.1 NAT 2015 1M

The figure shows a half-wave rectifier circuit with input voltage $v(t) = 10\sin(100\pi t)$ volts. Assuming ideal diode characteristic with zero forward voltage drop and zero reverse current, the average power consumed in watts by the load resistance R_L is _____ W.

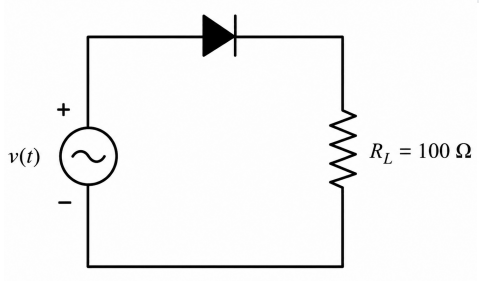


Fig. Q1.15.1

Q1.14.1 NAT 2014 1M

For the circuit shown in the figure assume ideal diodes with zero forward resistance and zero forward voltage drop. The current through the diode D_2 in mA is _____ .

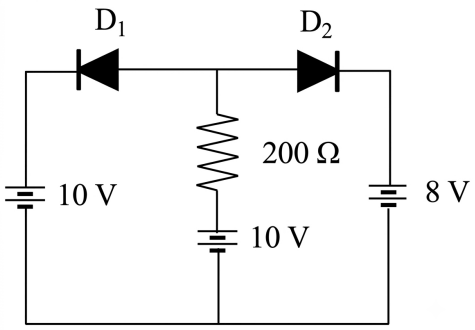


Fig. Q1.14.1

Q1.13.1 MCQ 2013 2M

A voltage $1000\sin\omega t$ volts is applied across YZ. Assuming ideal diodes, the voltage measured across WX in volts, is

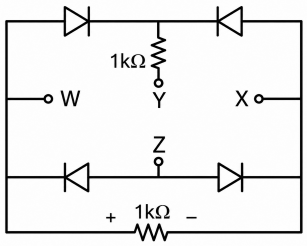


Fig. Q1.13.1

A. $\sin\omega t$ **B.** $(\sin\omega t + |\sin\omega t|)/2$

C. $(\sin\omega t - |\sin\omega t|)/2$ **D.** 0 for all t

Q1.13.2 MCQ 2013 2M

In the circuit shown below, the knee current of the ideal Zener diode is 10mA. To maintain 5 V across R_L, the minimum value of R_L in Ω and the minimum power rating of the Zener diode in mW respectively are

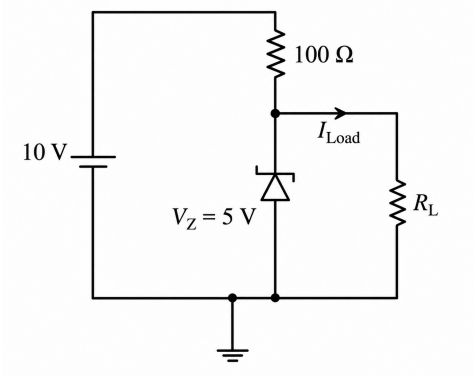


Fig. Q1.13.2

A. 125 and 125 **B.** 125 and 250
C. 250 and 125 **D.** 250 and 250

Q1.12.1 MCQ 2012 1M

The i-v characteristics of the diode in the circuit given below are

$$i = \begin{cases} \dfrac{v - 0.7}{500}\ \text{A}, & v \geq 0.7\text{V} \\ 0\ \text{A}, & v < 0.7\text{V} \end{cases}$$

The current in the circuit is

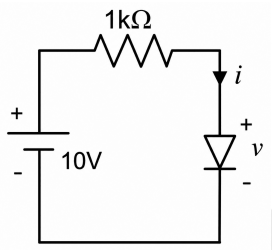


Fig. Q1.12.1

A. 10mA **B.** 9.3mA **C.** 6.67mA **D.** 6.2mA

Q1.11.1 MCQ 2011 2M

Assuming Zener diode D_1 has current-voltage characteristics as shown below on the right and forward voltage drop of diode D_2 is 0.7 V, the voltage V_0 in the circuit shown below is

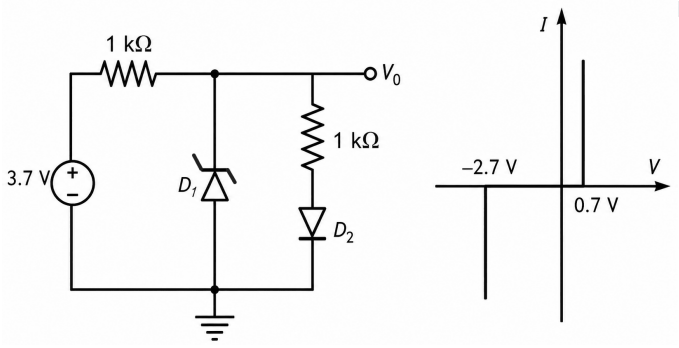


Fig. Q1.11.1

A. 3.7 V **B.** 2.7 V **C.** 2.2 V **D.** 0 V

Q1.08.1 MCQ 2008 1M

In the circuit shown below, the ideality factor η of the diode is unity and the voltage drop across it is 0.7 V. The dynamic resistance of the diode at room temperature is approximately

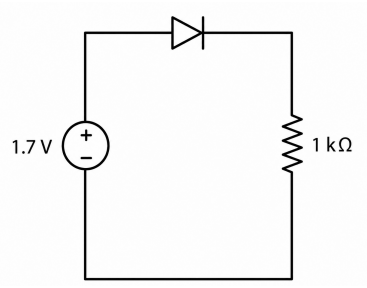


Fig. Q1.08.1

A. 15 Ω **B.** 25 Ω **C.** 50 Ω **D.** 700 Ω

Q1.05.1 MCQ 2005 1M

The peak value of the output voltage V_0 across the capacitor shown in the figure for a 230 : 9 transformer and a 230 V, 50 Hz, input, assuming 0.7 V diode drop and an ideal transformer, is

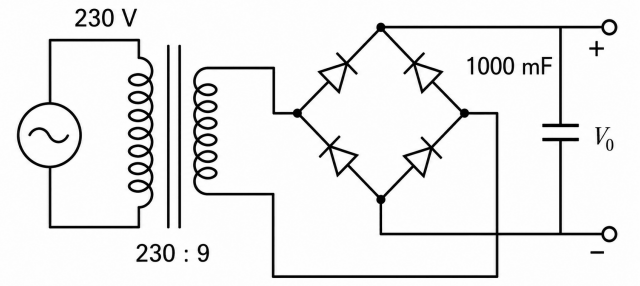


Fig. Q1.05.1

A. 12.73 **B.** 11.33 **C.** 7.6 **D.** 9.0

Q1.04.1 MCQ 2004 2M

For the circuit shown in the figure, the diode D is ideal. The power dissipated by the 300 Ω resistor is

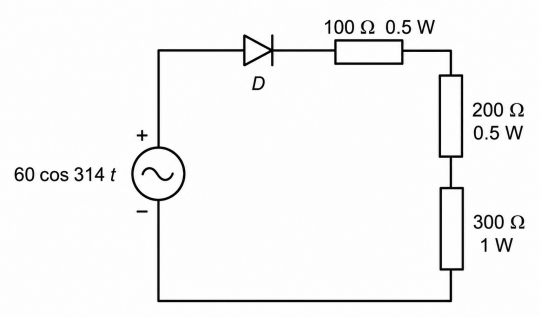


Fig. Q1.04.1

A. 0.25 W **B.** 0.50 W **C.** 0.75 W **D.** 1.00 W

Q1.04.2 MCQ 2004 1M

Assuming ideal diode characteristics, the input-output voltage relationship for the circuit shown in figure is

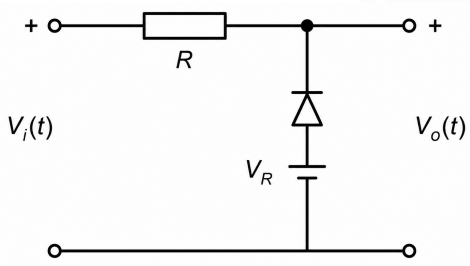


Fig. Q1.04.2

A. $V_o(t) = V_i(t)$, for all $V_i(t)$

B. $V_o(t) = V_i(t)$, for $V_i(t) > V_R$
$V_o(t) = 0$, otherwise

C. $V_o(t) = V_i(t)$, for $V_i(t) < V_R$
$V_o(t) = V_R$, otherwise

D. $V_o(t) = V_i(t)$, for $V_i(t) > V_R$
$V_o(t) = V_R$, otherwise

Q1.03.1 MCQ 2003 2M

The 5 V Zener diode in figure is ideal and the ammeter (A), of full-scale 1 mA, has an internal resistance of 100 Ω. The circuit shown, with terminal 1 positive, functions as a

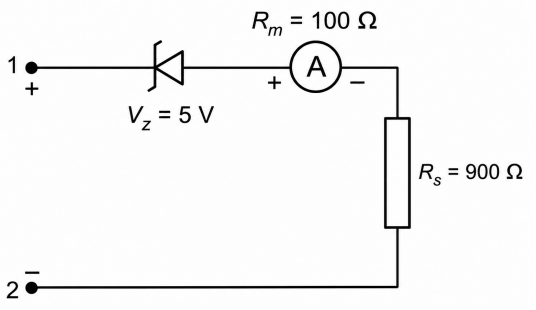


Fig. Q1.03.1

A. 0 – 1 V DC voltmeter **B.** 0 – 1 mA DC ammeter
C. 0 – 6 V DC voltmeter **D.** 0 – 5 V AC voltmeter

Q1.03.2 MCQ 2003 2M

A forward-biased silicon diode when carrying negligible current, has a voltage drop of 0.64 V. When the current is 1 A it dissipates 1 W. The ON-resistance of the diode is

A. 0.36 Ω **B.** 0.64 Ω **C.** 0.74 Ω **D.** 1.0 Ω

Q1.00.1 MCQ 2000 1M

For a sinusoidal input of 50 V amplitude, the circuit shown in figure can be used as

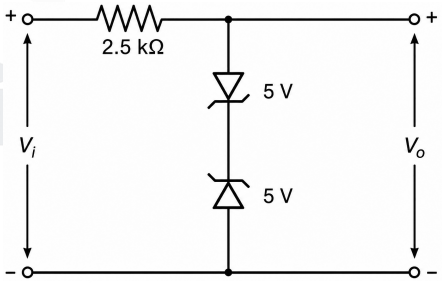


Fig. Q1.00.1

A. Regulated DC power supply

B. Square Wave generator

C. Half wave rectifier

D. Full wave rectifier

Q1.99.1 MCQ 1999 1M

The input voltage, V_i is $4+3\sin\omega t$. Assuming all elements to be ideal. The average of the output voltage V_0 in figure is

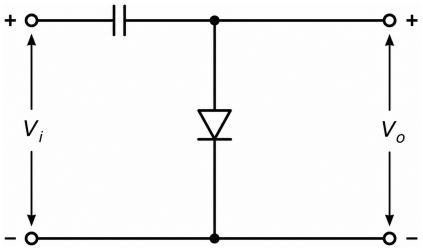


Fig. Q1.99.1

A. –3 V **B.** +3 V **C.** –7 V **D.** +7 V

Q1.99.2 MCQ 1999 1M

In the circuit shown in figure, V_i is 4 V. Assuming the diodes to be ideal, V_0 is

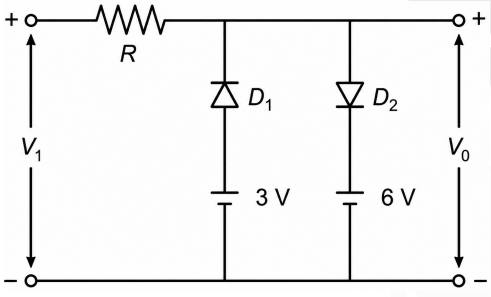


Fig. Q1.99.2

A. 3 V **B.** 4 V **C.** 4.5 V **D.** 6 V

PrepFusion

Q1.08.1 B	Q1.05.1 B	Q1.04.1 C
Q1.04.2 D	Q1.03.1 C	Q1.03.2 A
Q1.00.1 B	Q1.99.1 A	Q1.99.2 B

Answer Key

Q1.26.1 11 to 11	Q1.23.1 B	Q1.23.2 0.95 to 1.05
Q1.22.1 6.0 to 6.0	Q1.22.2 D	Q1.22.3 -10.1 to -9.9
Q1.20.1 A	Q1.20.2 1.8 to 1.8	Q1.19.1 B
Q1.18.1 200 to 200	Q1.17.1 D	Q1.16.1 0 to 0
Q1.15.1 0.24 to 0.26	Q1.14.1 10 to 10	Q1.13.1 D
Q1.13.2 B	Q1.12.1 D	Q1.11.1 C

UNIT

II

MOSFETs

Topics

1. Introduction to MOS - Depletion and Enhancement Types, MOS Physics
2. Biasing a MOSFET - Transconductance, Output Impedance and Small Signal Analysis
3. Small Signal Model of MOSFET and Port Impedance
4. Common Source Amplifier - With and Without Degeneration, Different Biasing
5. Miller's Theorem and GmRout Method for Gain
6. Common Gate, Common Drain and PMOS Amplifiers
7. Cascode Current Source, Amplifiers and Current Mirror Circuits
8. Multistage Amplifiers and Pass Transistor Logic
9. Frequency Response of MOSFET and Common Gate Amplifiers
10. Introduction to Differential Amplifiers, Current Source, Voltage Source as Tail
11. Large Signal Analysis of Differential Amplifiers, Concept of Half Circuit
12. Building a Two-Stage Op Amp
13. Types and Effects of Feedback
14. Feedback Topologies - Voltage-Voltage, Voltage-Current, Current-Voltage, Current-Current
15. MOS Amplifiers with Body Effect

Unit Snapshot*

Stream: IN	**Questions:** 13	**MCQ:** 7	**MSQ:** 0	**NAT:** 6
1M: 3	**2M:** 10	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2026)–4M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M

A common gate amplifier represents a _____.

A. voltage controlled voltage source

B. current controlled current source

C. voltage controlled current source

D. current controlled voltage source

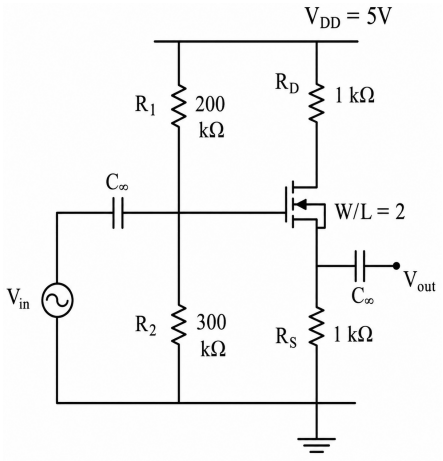


Fig. Q2.23.1

Q2.24.1 NAT 2024 1M

Figure shows an amplifier using an NMOS transistor. Assume that the transistor is in saturation with device parameters, $\mu_n C_{ox} = 250\ \mu A/V^2$, threshold voltage $V_T = 0.65$ V and $W/L = 4$. Ignore the channel length modulation effect. The drain current of the transistor at the operating point is _____ μA **(rounded off to nearest integer)**.

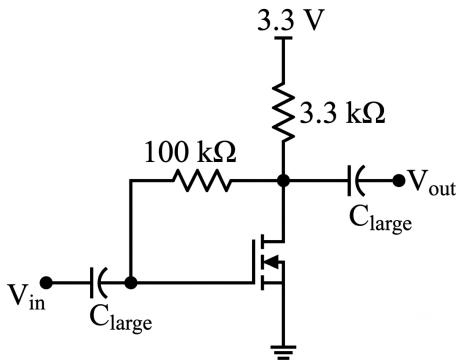


Fig. Q2.24.1

Q2.23.1 NAT 2023 2M

The figure below shows a feedback amplifier constructed using an nMOS transistor. Assume that $\mu_n C_{ox} = 1$ mA/V^2, threshold voltage $V_T = 1$ V and $W/L = 2$. The bias voltage at the drain terminal is 4 V. The capacitors C_∞ offer zero impedance at the signal frequency. The ratio V_{out}/V_{in} is _____ **(rounded off to two decimal places)**.

Q2.22.1 MCQ 2022 2M

In the small signal circuit shown, the enhancement mode n-channel MOSFET is biased in saturation with transconductance g_m. If channel length modulation is ignored, the small signal impedance looking into the node P is given by _____ .

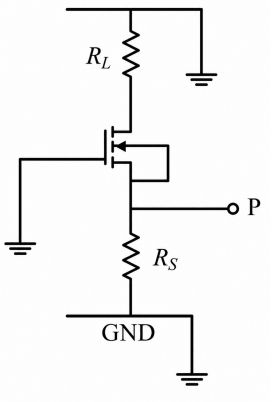


Fig. Q2.22.1

A. $R_S \parallel R_L \parallel g_m^{-1}$

B. $R_S \parallel g_m^{-1}$

C. $(R_S + R_L) \parallel g_m^{-1}$

D. $\dfrac{R_L g_m}{1 + R_S g_m}(R_L \parallel g_m^{-1})$

Q2.22.2 MCQ 2022 2M

In the small signal circuit shown, the enhancement mode n-channel MOSFET is biased in saturation with a transconductance g_m. A small signal low-frequency voltage v_d injected at the supply terminal results in a small signal voltage fluctuation v_o at the output. If the channel length modulation of the MOSFET is ignored, the small signal gain v_o/v_d is given by _____ .

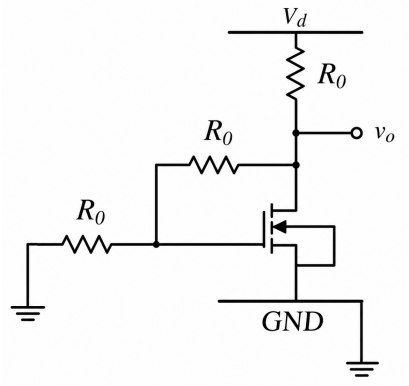


Fig. Q2.22.2

A. $\dfrac{-g_m R_0}{1 + g_m R_0}$

B. $(g_m R_0 + 1)^{-1}$

C. $\dfrac{-g_m R_0}{1 + 2g_m R_0}$

D. $\left(\dfrac{g_m R_0}{2} + \dfrac{3}{2}\right)^{-1}$

Q2.19.1 MCQ 2019 2M

A voltage amplifier is constructed using enhancement mode MOSFETs labeled M1, M2, M3 and M4 in the figure below. M1, M2 and M4 are n-channel MOSFETs and M3 is a p-channel MOSFET. All MOSFETs operate in saturation mode and channel length modulation can be ignored. The low frequency, small signal input and output voltages are v_{in} and v_{out} respectively and the dc power supply voltage is V_{DD} . All n-channel MOSFETs have identical transconductance g_{mn} while the p-channel MOSFET has transconductance g_{mp}. The expressions for the low frequency small signal voltage gain v_{out}/v_{in} is

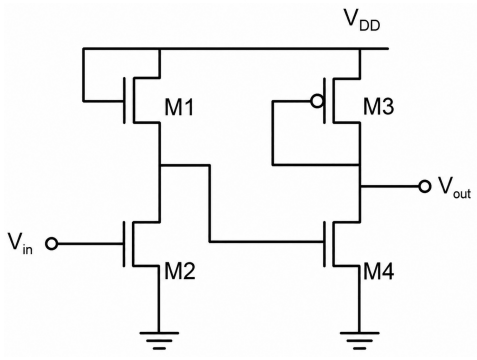


Fig. Q2.19.1

A. $-g_{mn}/g_{mp}$

B. $-g_{mn}(g_{mn} + g_{mp})^{-1}$

C. $+g_{mn}/g_{mp}$

D. $g_{mn}(g_{mn} + g_{mp})^{-1}$

Q2.19.2 NAT 2019 2M

In the circuit shown below, all transistors are n-channel enhancement mode MOSFETs. They are identical and are biased to operate in saturation mode. Ignoring channel length modulation, the output voltage V_{out} is _____ V.

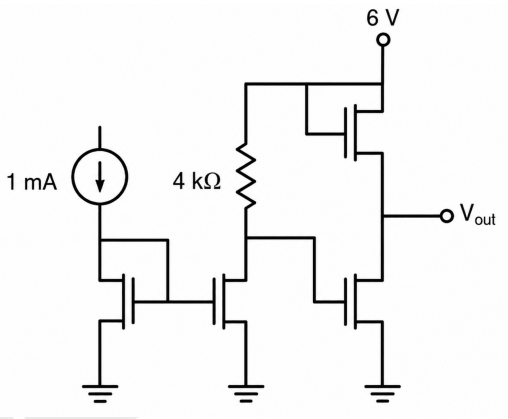


Fig. Q2.19.2

Q2.17.1 NAT 2017 2M

In the circuit, shown in the figure, the MOSFET is operating in the saturation zone. The characteristics of the MOSFET is given by

$$I_D = \frac{1}{2}(V_{GS} - 1)^2 \text{ mA, where } V_{GS} \text{ is in V.}$$

If $V_s = +5$ V, then the value of R_s in kΩ is _____ .

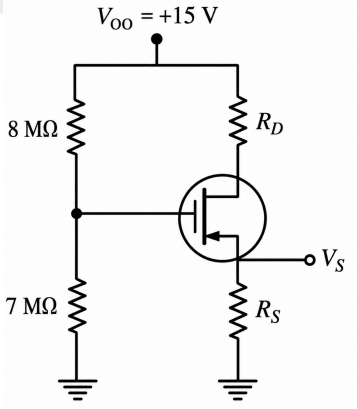


Fig. Q2.17.1

Q2.15.1 NAT 2015 2M

In the circuit shown in the figure, both the NMOS transistors are identical with their threshold voltages being 5 V. Ignoring channel length modulation, the output voltage V_{out} in volt is _____ V.

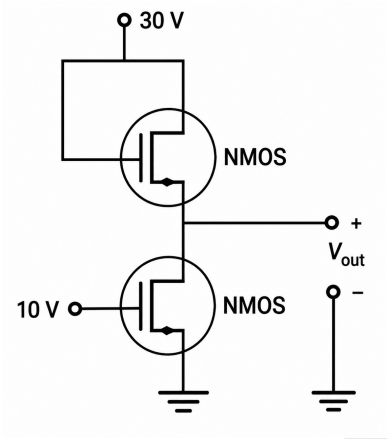


Fig. Q2.15.1

Q2.15.2 NAT 2015 2M

In the circuit shown, the voltage source $v(t) = 15 + 0.1\sin(100t)$. The PMOS transistor is biased such that it is in saturation with its gate-source capacitance being 4 nF and its transconductance at the operating point being 1 mA/V. Other parasitic impedances of the MOSFET may be ignored. An external capacitor of capacitance 2 nF is connected across the PMOS transistor as shown. The input impedance in mega ohm as seen by the voltage source is _____ MΩ.

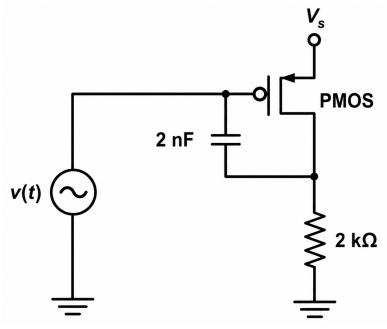


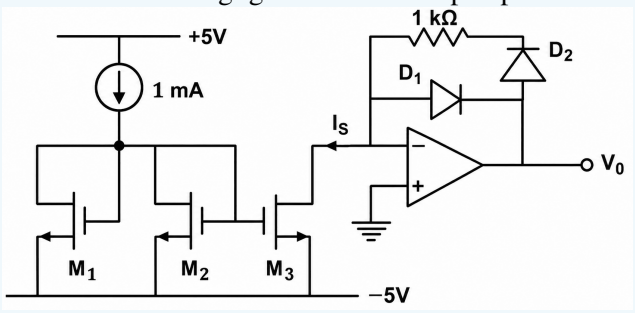
Fig. Q2.15.2

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q2.11.1 and Q2.11.2

M1, M2 and M3 in the circuit shown below are matched N-channel enhancement mode MOSFETs operating in saturation mode, forward voltage drop of each diode is 0.7 V, reverse leakage current of each diode is negligible and the opamp is ideal.



Q2.11.1 MCQ 2011 2M

The current I_S in the circuit is

A. –1 mA **B.** 0.5 mA **C.** 1 mA **D.** 2 mA

Q2.11.2 MCQ 2011 2 M

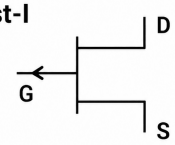
For the computed value of current I_S, the output voltage V_0 is

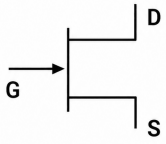
A. 1.2V **B.** 0.7V **C.** 0.2V **D.** –0.7V

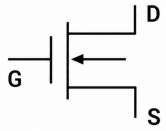
Q2.98.1 MCQ 1998 1M

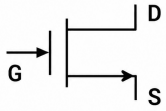
Match the figures with the devices names

List-I

A. 

B. 

C. 

D. 

List-II

1. Depletion mode MOSFET
2. P-channel JFET
3. N-channel JFET
4. Enhancement Mode MOSFET

A.

A	B	C	D
2	3	4	1

B.

A	B	C	D
1	2	3	4

C.

A	B	C	D
1	3	4	2

D.

A	B	C	D
3	1	2	4

Answer Key

Q2.26.1 B	Q2.24.1 498 to 502	Q2.23.1 0.64 to 0.7
Q2.22.1 B	Q2.22.2 D	Q2.19.1 C
Q2.19.2 4	Q2.17.1 9.9 to 10.1	Q2.15.1 20
Q2.15.2 1	Q2.11.1 B	Q2.11.2 A
Q2.98.1 A		

UNIT

III

Operational Amplifier

Topics

1. Basics of Op-Amp, Comparator and Virtual Short
2. Inverting and Non-Inverting Amplifiers
3. Adder, Differential and Instrumentation Amplifiers
4. Log, Anti-Log and Rectifier Using Op-Amp
5. Filters Using Op-Amp
6. Offsets in Op-Amp
7. CMRR, Slew Rate and Frequency Response of Op-Amp
8. Integrator Using Op-Amp
9. Differentiator Using Op-Amp
10. Active Clipper, Clamper and Peak Detector
11. Multivibrators Using Op-Amp
12. Oscillators Using Op-Amp
13. Schmitt Triggers

Unit Snapshot*

Stream: IN	**Questions:** 127	**MCQ:** 86	**MSQ:** 1	**NAT:** 40
1M: 49	**2M:** 78	**Descriptive:** 0	**Avg Weightage** ~7M	**Range:** 2M(2025)–14M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M

In the circuit shown below, R is 1 kΩ and R_f is 10 kΩ. If V_{in} is 100 mV and the op-amps have supply voltages of ±15 V, then V_{out} is _____ V.

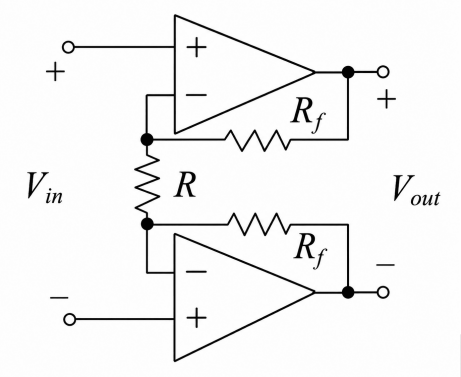


Fig. Q3.26.1

A. −15 **B.** 15 **C.** 2.1 **D.** 1.1

Q3.26.2 MCQ 2026 2M

In the following active filter circuit with ideal op-amps, R is 1 kΩ, C is 1 μF, and V_{in} has an amplitude of 1 V at 1000 rad/s.

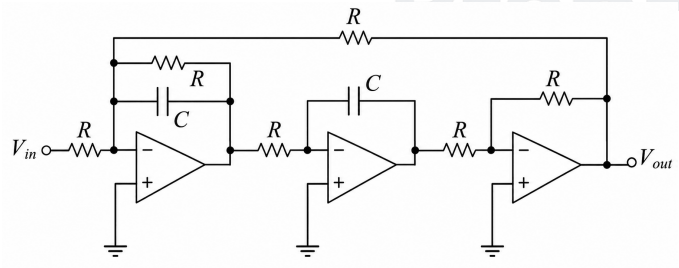


Fig. Q3.26.2

The amplitude of V_{out} is _____ V.

A. 0.5 **B.** 0.707 **C.** 1.0 **D.** 1.414

Q3.25.1 MCQ 2025 1M

For the ideal opamp based circuit shown in the figure, the ratio $\frac{V}{I}$ is

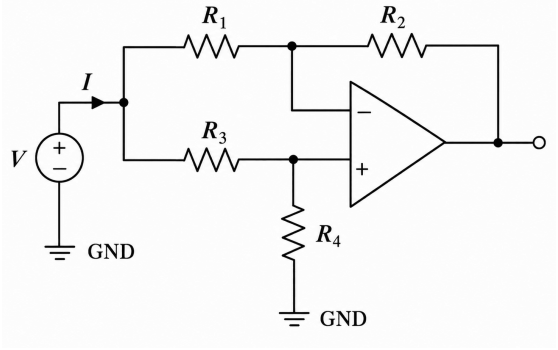


Fig. Q3.25.1

A. $\left(\frac{R_3 + R_4}{R_1 + R_3}\right) R_1$ **B.** $\left(\frac{R_2 + R_4}{R_1 + R_3}\right) R_3$

C. $R_1 + R_3$ **D.** $R_3 + R_4$

Q3.25.2 MSQ 2025 1M

The circuits mentioned in the following options are realized using ideal opamp. Among these, the circuit(s) performing non-linear operation on the input signal is/are

- ☐ **A.** Instrumentation amplifier
- ☐ **B.** Schmitt trigger
- ☐ **C.** Logarithmic amplifier
- ☐ **D.** Precision rectifier

Q3.24.1 NAT 2024 1M

Consider the circuit shown in the figure.

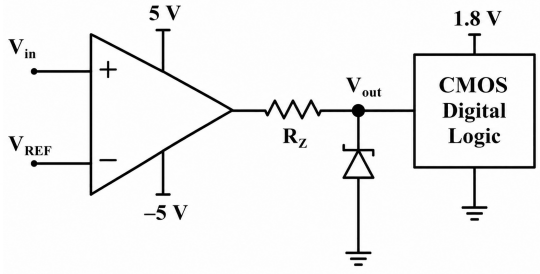


Fig. Q3.24.1

The CMOS digital logic circuit has infinite input impedance. Assume the opamp is ideal. A 1.8 V Zener diode with a minimum Zener current of 2 mA is used. The corresponding maximum value of resistance R_Z is _____ kΩ. **(rounded off to one decimal place)**.

Q3.24.2 NAT 2024 1M

Consider the readout circuit of a piezoelectric sensor shown in the figure.

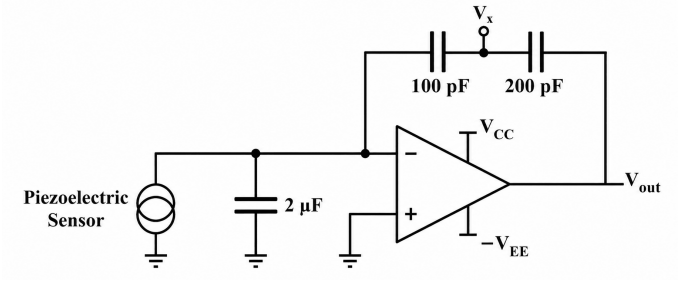


Fig. Q3.24.2

When the piezoelectric sensor generates a charge q_p, the resulting change in voltage V_x is −2 V. Then the corresponding change in the voltage V_{out} is _____ V **(rounded off to nearest integer)**.

Note: Assume all components are ideal.

Q3.24.3 MCQ 2024 2M

In the figure shown, both the opamps A_1 and A_2 are ideal, except that the opamp A_1 has an offset voltage (V_{os}) of 1 mV. For V_{in} = 0 V, the values of the output voltages V_{out1} and V_{out2}, respectively, are

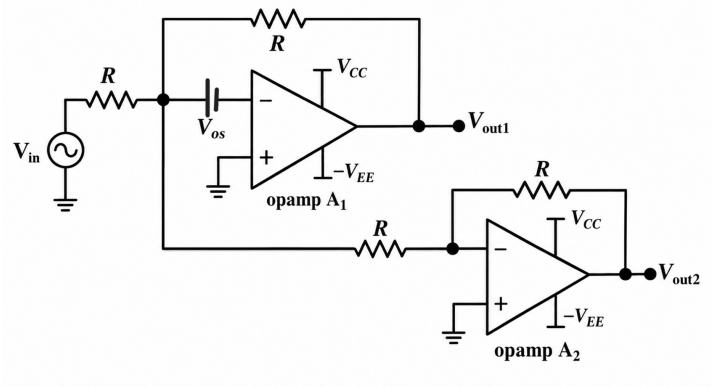


Fig. Q3.24.3

A. 3 mV and −1 mV
B. 1 mV and 0 mV
C. 1 mV and −1 mV
D. 2 mV and 0 mV

Q3.24.4 MCQ 2024 2M

In the figure shown, R = 1 kΩ and C = 0.1 μF. For a dc gain of −10, the 3 dB cut-off frequency **(rounded off to one decimal place)** is

Assume the opamp is ideal.

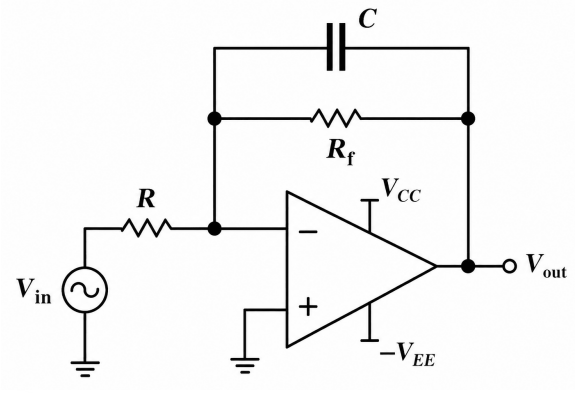


Fig. Q3.24.4

A. 159.1 Hz
B. 1591.5 Hz
C. 1750.7 Hz
D. 175.0 Hz

Q3.24.5 NAT 2024 2M

In the figure shown, R = 4.5 kΩ, ΔR = 1.5 kΩ, and INA is assumed to be ideal. The equivalent resistance between A and B is _____ kΩ **(rounded off to nearest integer)**.

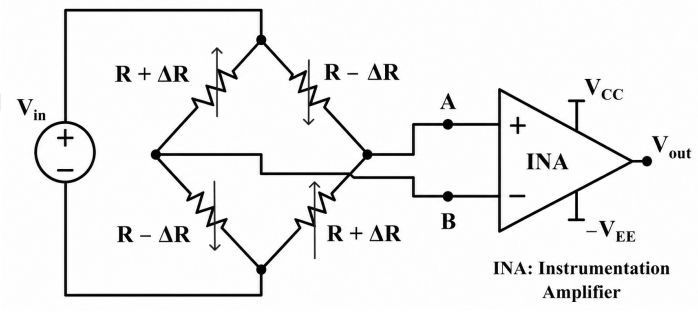


Fig. Q3.24.5

Q3.24.6 NAT 2024 2M

Consider the capacitive sensor circuit and its output voltage shown in the figure. The circuit is switched ON at t = 0. Assuming the opamp to be ideal, the frequency of the output voltage V_o is _____ kHz **(rounded off to two decimal places)**.

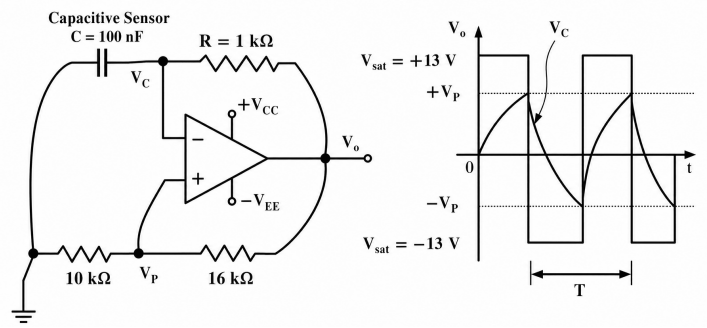


Fig. Q3.24.6

Q3.24.7 NAT 2024 2M

In the figure shown, the diode current is given by $I_D = I_S e^{\frac{\alpha V_D}{T}}$. V_D is the diode voltage in volts, T is the absolute temperature in Kelvin, $\alpha = 1.16 \times 10^4$ K/V, and $I_S = 10^{-15}$ A is the saturation current. The dc current source, opamp and the resistors are ideal, and are assumed to be temperature independent. The change in the output voltage (V_{out}) per Kelvin change in temperature is _____ mV **(rounded off to one decimal place)**.

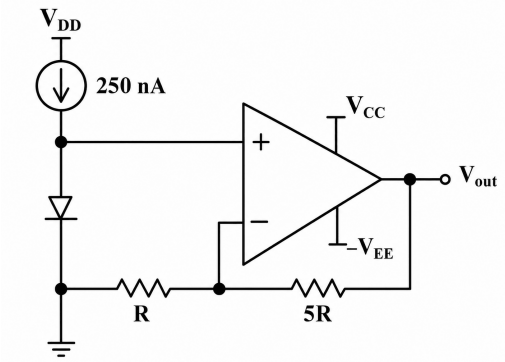


Fig. Q3.24.7

Q3.24.8 NAT 2024 2M

In the circuit shown, assume the opamp is ideal and the initial charge on the capacitor is zero. The output voltage at time $t = 2$ ms is _____ V **(rounded off to one decimal place)**.

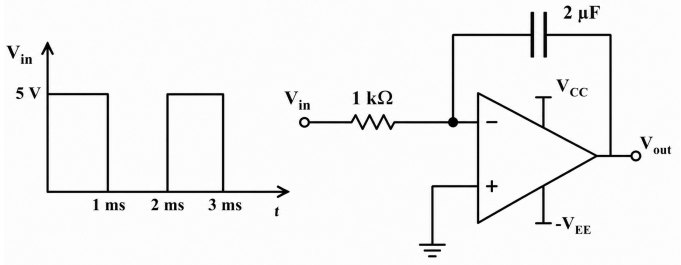


Fig. Q3.24.8

Q3.23.1 NAT 2023 1M

The opamp in the circuit shown is ideal, except that it has an input bias current of 1 nA and an input offset voltage of 10 μV. The resulting worst-case output voltage will be $\pm$_____ μV **(rounded off to the nearest integer)**.

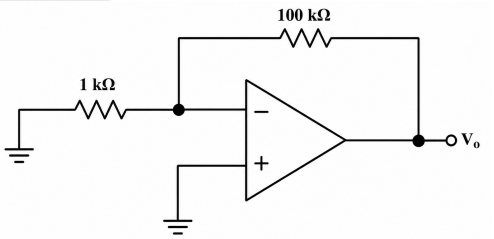


Fig. Q3.23.1

Q3.23.2 NAT 2023 2M

The opamps in the circuit are ideal. The input signals are $V_{S1} = 3+0.10\sin(300t)$ V and $V_{S2} = -2+0.11\sin(300t)$ V. The average value of the voltage V_0 is _____ V **(rounded off to two decimal places)**.

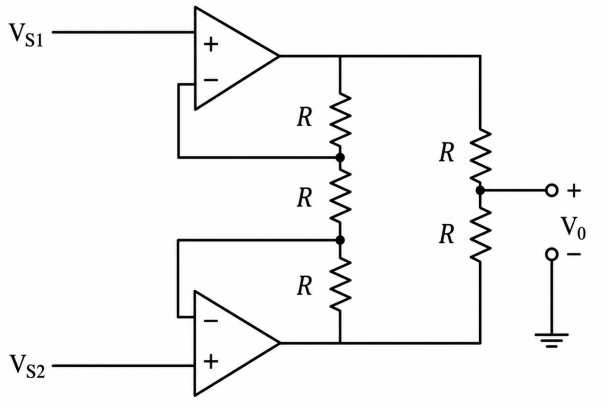


Fig. Q3.23.2

Q3.23.3 NAT 2023 2M

In the circuit shown, the input voltage V_{in} = 100 mV. The switch and the opamp are ideal. At time t = 0, the initial charge stored in the 10 nF capacitor is 1 nC, with the polarity as indicated in the figure. The switch S is controlled using a 1 kHz square-wave voltage signal V_s as shown. Whenever V_s is 'High', S is in position '1' and when V_s is 'Low', S is in position '2'.

At t = 20 ms, the magnitude of the voltage V_o will be _____ mV **(rounded off to the nearest integer)**.

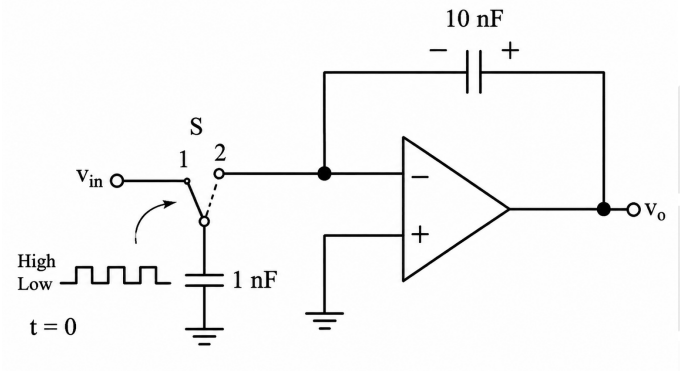


Fig. Q3.23.3

Q3.23.4 NAT 2023 2M

In the circuit shown, assuming an ideal opamp, the value of the output voltage V_o = _____ V **(rounded off to one decimal place)**.

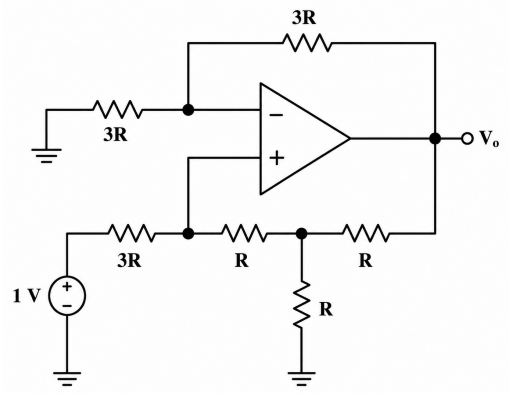


Fig. Q3.23.4

Q3.22.1 NAT 2022 1M

An OPAMP has a gain of 10^4, an input impedance of 10 MΩ and an output impedance of 100 Ω. The OPAMP is used in unity-gain feedback configuration in a voltage buffer circuit. The closed-loop output impedance of the OPAMP (in milliohms) in the circuit is _____ **(round off to one decimal place)**.

Q3.22.2 NAT 2022 2M

The circuit shown is driven by a sinusoidal input voltage, V_{in}, resulting in the output voltage, V_{out}. The frequency (in kilohertz) at which the voltage gain is 0 dB is _____ **(round off to two decimal places)**.

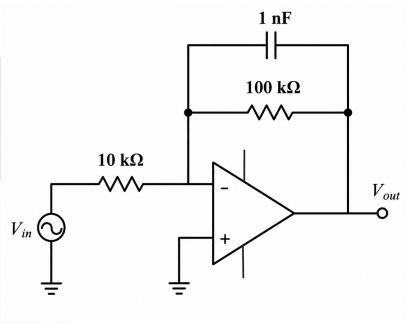


Fig. Q3.22.2

Q3.22.3 NAT 2022 2M

In the circuit shown, the switch is initially closed. It is opened at t = 0 s and remains open thereafter. The time (in milliseconds) at which the output voltage V_{out} becomes LOW is _____ **(round off to three decimal places)**.

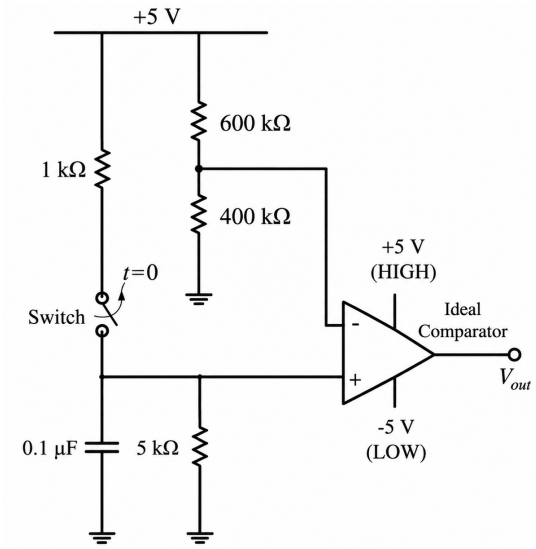


Fig. Q3.22.3

Q3.21.1 NAT 2021 1M

The diode used in the circuit has a fixed voltage drop of 0.6 V when forward biased. A signal v_s is given to the ideal OpAmp as shown. When v_s is at its positive peak, the output (v_{OA}) of the OpAmp in volts is _____ .

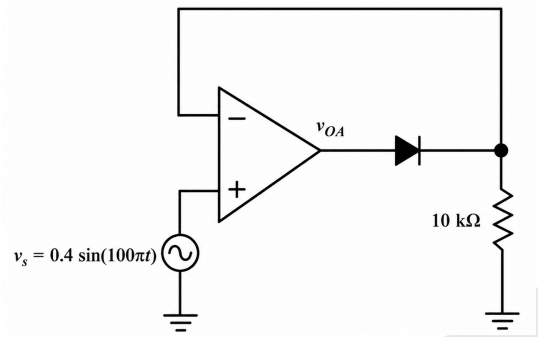


Fig. Q3.21.1

Q3.21.2 MCQ 2021 2M

The output V_o of the ideal OpAmp used in the circuit shown below is 5 V. Then the value of resistor R_L in kilo ohm (kΩ) is

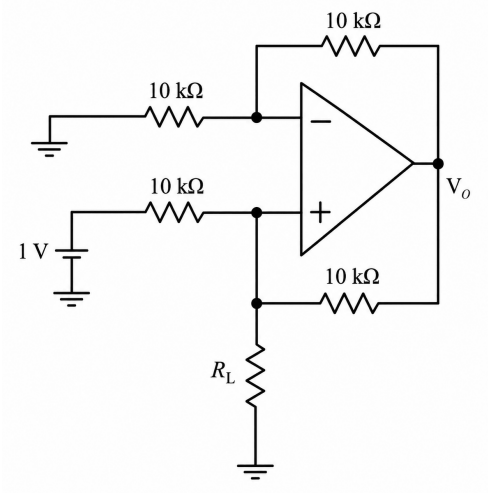


Fig. Q3.21.2

A. 2.5 **B.** 5 **C.** 25 **D.** 5

Q3.21.3 NAT 2021 2M

The circuit shown below uses an ideal OpAmp. Output V_O in volt is _____ **(rounded off to one decimal place)**.

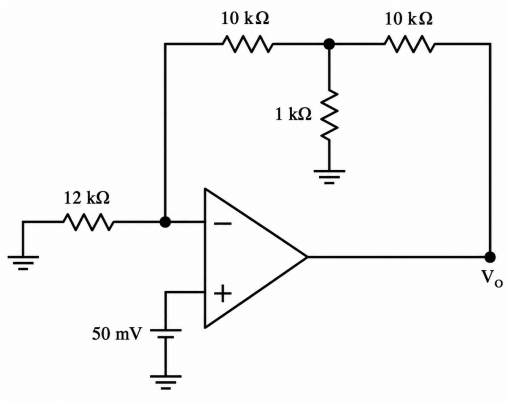


Fig. Q3.21.3

Q3.20.1 NAT 2020 1M

Assuming ideal opamps, the output voltage at V_1 in the figure shown **(in volts)** is _____ .

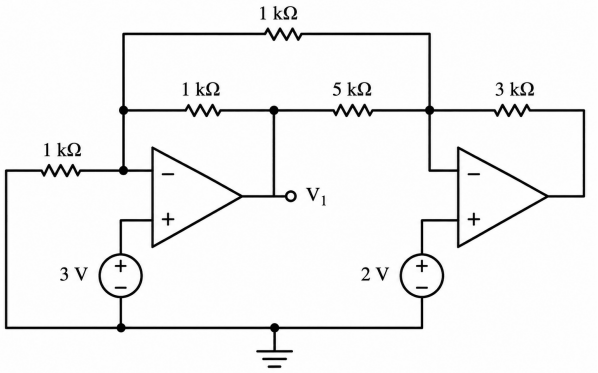


Fig. Q3.20.1

Q3.20.2 NAT 2020 1M

Assume that the opamp in the circuit shown is ideal. The value of $\frac{V_x}{I_x}$ **(in kΩ)** is _____ .

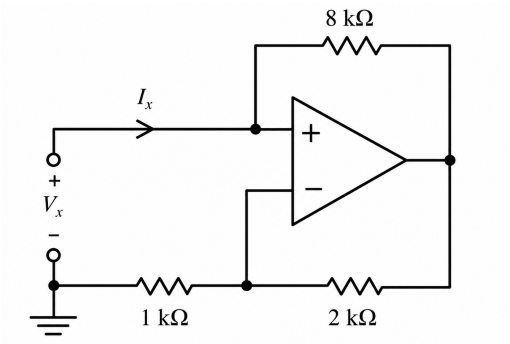


Fig. Q3.20.2

Q3.20.3 NAT 2020 2M

Assuming that the opamp used in the circuit shown is ideal, the reading of the 1 Hz bandwidth, permanent magnet moving coil (PMMC) type voltmeter **(in volts)** is _____ .

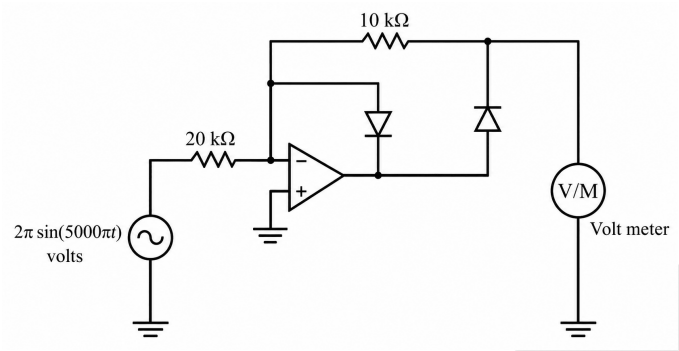


Fig. Q3.20.3

Q3.20.4 NAT 2020 2M

If the opamps in the circuit shown are ideal and V_x = 0.5 mV, the steady state value of V_O **(in volts, rounded off to two decimal places)** is _____ .

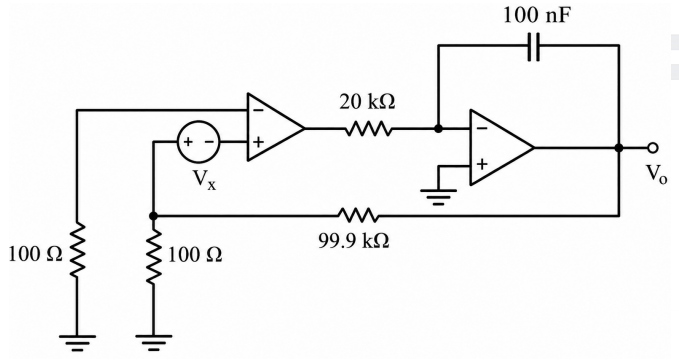


Fig. Q3.20.4

Q3.20.5 NAT 2020 2M

The circuit shown uses ideal opamp powered from a supply V_{CC} = 5 V. If the charge q_p generated by the piezoelectric sensor is of the form $q_p = 0.1\sin(10000\pi t)$ μC, the peak detector output after 10 cycles of q_p **(in volts, rounded off to one decimal place)** is _____ .

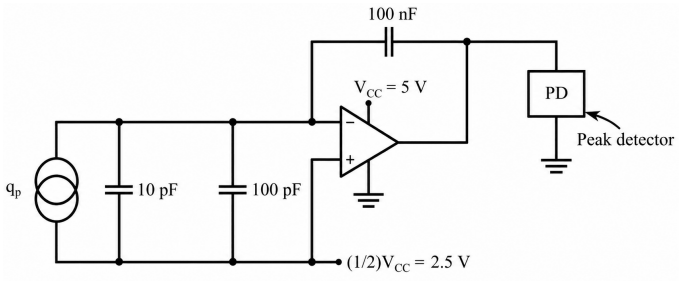


Fig. Q3.20.5

Q3.19.1 NAT 2019 2M

In the circuit shown below all OPAMPS are ideal. The current I = 0 A when the resistance R = _____ kΩ.

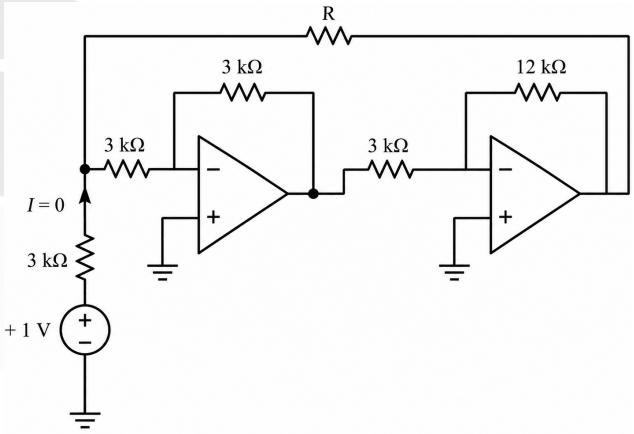


Fig. Q3.19.1

Q3.19.2 MCQ 2019 1M

In the circuit shown below, the input voltage V_{in} is positive. The current (I)-voltage (V) characteristics of the diode can be assumed to be $I = I_0 e^{V/V_T}$ under the forward bias condition, where V_T is the thermal voltage and I_0 is the reverse saturation current. Assuming an ideal op-amp, the output voltage V_{out} of the circuit is proportional to

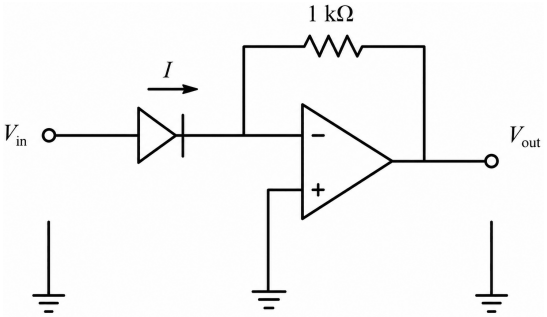


Fig. Q3.19.2

A. $\log_e\left(\frac{V_{in}}{V_T}\right)$ **B.** $2V_{in}$

C. e^{V_{in}/V_T} **D.** V_{in}^2

Q3.18.1 NAT 2018 2M

The circuit given use ideal opamps. The current I (in μA) drawn from the source v_s is (up to two decimal places) _____ .

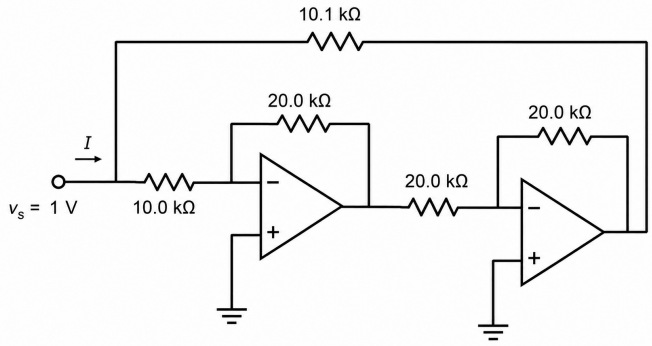


Fig. Q3.18.1

Q3.18.2 NAT 2018 2M

An opamp that is powered from a ±5 V supply is used to build a non-inverting amplifier having a gain of 15. The slew rate of the opamp is 0.5×10^6 V/s. For a sinusoidal input with amplitude of 0.3 V, the maximum frequency (in kHz) up to which it can be operated without any distortion is (up to one decimal place) _____.

Q3.18.3 NAT 2018 2M

In the given relaxation oscillator, the opamps and the zener diodes are ideal. The frequency (in kHz) of the square wave output v_0 is _____.

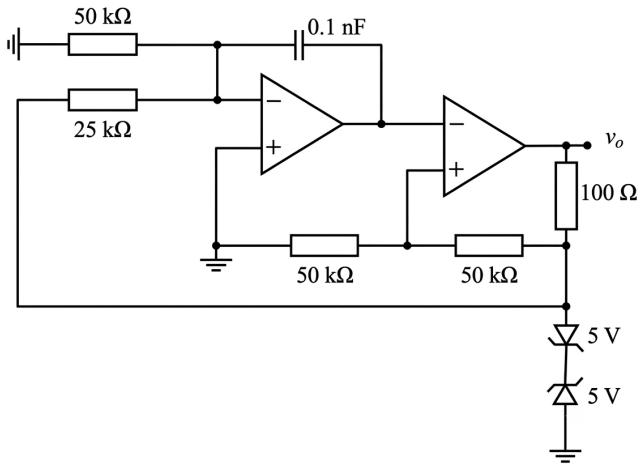


Fig. Q3.18.3

Q3.17.1 NAT 2017 2M

The two-input voltage multiplier, shown in the figure has a scaling factor of 1 and produces voltage output. If $V_1 = +15$ V and $V_2 = +3$ V, the value of the V_0 in volt is _____ .

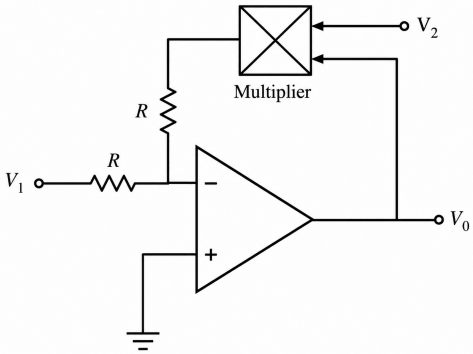


Fig. Q3.17.1

Q3.17.2 NAT 2017 1M

The differential amplifier, shown in the figure, has a differential gain of $A_d = 100$ and common mode gain of $A_c = 0.1$. If $V_1 = 5.01$ V and $V_2 = 5.00$ V, then V_0 in volt (up to one decimal place) is _____ .

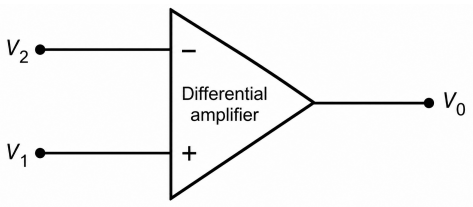


Fig. Q3.17.2

Q3.17.3 MCQ 2017 1M

The output V_0 shown in the figure, in volt, is close to

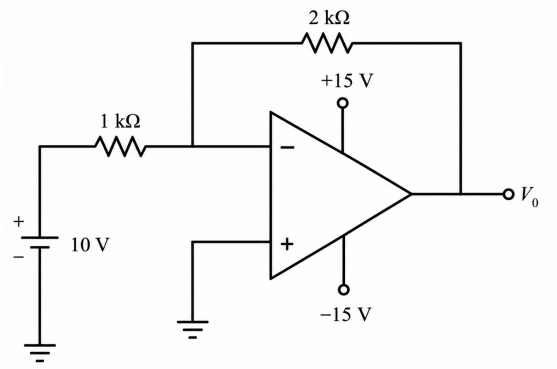


Fig. Q3.17.3

A. −20 **B.** −15 **C.** −5 **D.** 0

Q3.17.4 NAT 2017 2M

The circuit of a Schmitt trigger is shown in the figure. The Zener-diode combination maintains the output between ±7 V. The width of the hysteresis band is _____ V.

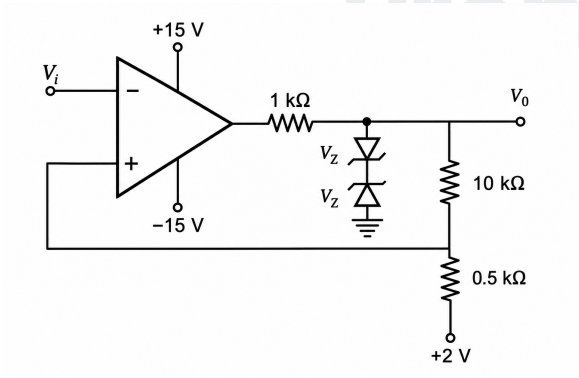


Fig. Q3.17.4

Q3.17.5 NAT 2017 2M

Assuming the opamp shown in the figure to be ideal, the frequency at which the magnitude of V_0 will be 95% of the magnitude of V_{in} is _____ kHz.

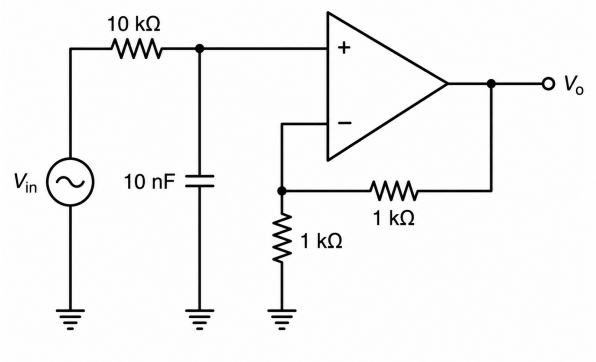


Fig. Q3.17.5

Q3.17.6 MCQ 2017 1M

The condition for oscillation in a feedback oscillator circuit is that at the frequency of oscillation, initially the loop gain is greater than unity while the total phase shift around the loop in degree is

A. 0 **B.** 90 **C.** 180 **D.** 270

Q3.16.1 NAT 2016 2M

In the circuit given below, the opamp is ideal. The value of current I_L in microampere is _____ .

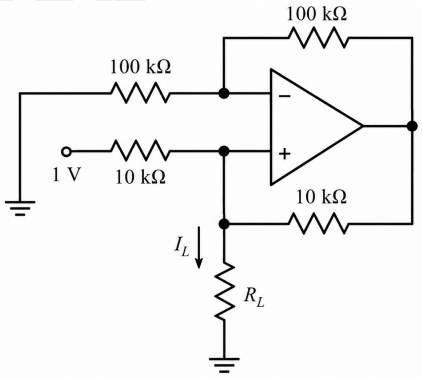


Fig. Q3.16.1

Q3.16.2 NAT 2016 2M

In the circuit given below, the opamp is ideal. The input v_x is a sinusoid. To ensure $v_y = v_x$, the value of C_N in Pico farad is _____ .

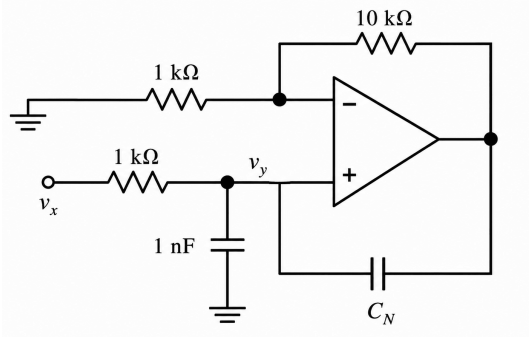


Fig. Q3.16.2

Q3.16.3 NAT | 2016 | 2M

An ideal opamp is used to realize a difference amplifier circuit given below having a gain of 10. If $x = 0.025$, the CMRR of the circuit in dB is _____ .

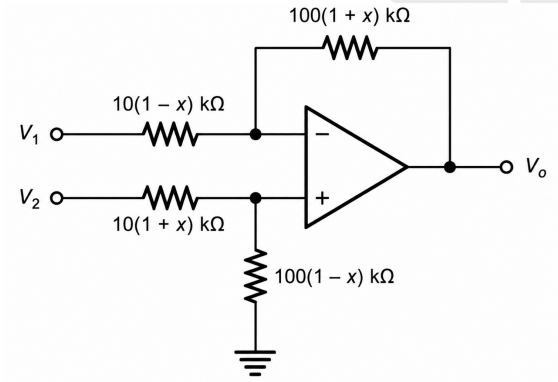


Fig. Q3.16.3

Q3.16.4 NAT | 2016 | 1M

In the circuit given below, each input terminal of the opamp draws a bias current of 10 nA. The effect due to these input bias currents on the output voltage V_0 will be zero, if the value of R chosen in kilo-ohm is _____ .

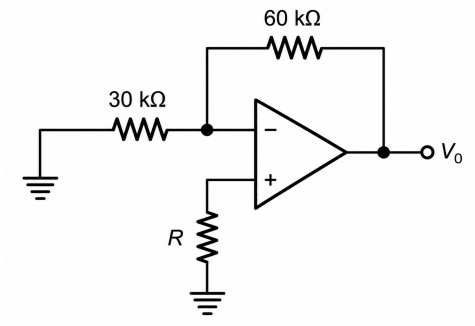


Fig. Q3.16.4

Q3.16.5 NAT | 2016 | 1M

In the circuit given below, the diodes D_1 and D_2 have a forward voltage drop of 0.6 V. The opamp used is ideal. The magnitude of the negative peak value of the output V_0 in volt is _____ .

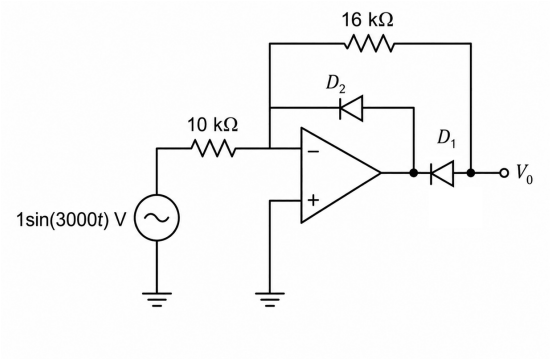


Fig. Q3.16.5

Q3.16.6 NAT | 2016 | 1M

In the circuit given below, the opamp is ideal. The output voltage V_o in volt is _____ .

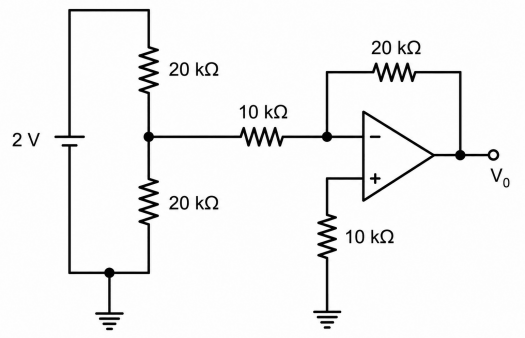


Fig. Q3.16.6

Q3.15.1 NAT | 2015 | 2M

In the figure shown, R_T represents a resistance temperature device (RTD), whose characteristic is given by $R_T = R_o(1 + \alpha T)$, where $R_o = 100\ \Omega$, $\alpha = 0.0039\ ^\circ C^{-1}$ and T denotes the temperature in °C. Assuming the opamp to be ideal, the value of V_o in volts when $T = 100\ ^\circ C$, is _____ V.

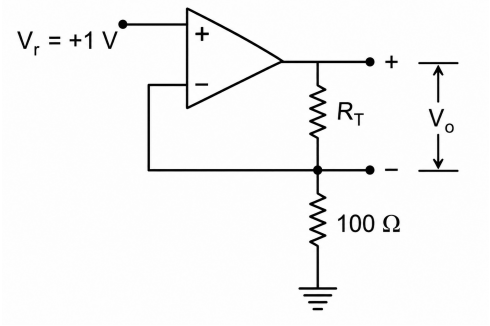


Fig. Q3.15.1

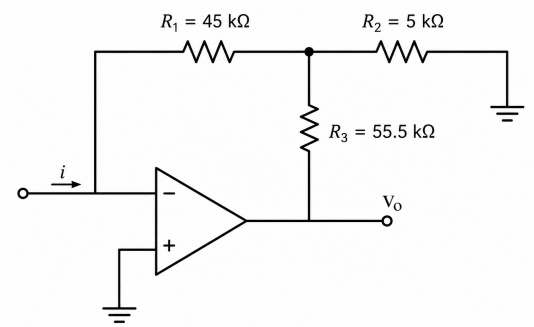


Fig. Q3.14.1

Q3.15.2 MCQ 2015 2M

An opamp has ideal characteristics except that its open loop gain is given by the expression $A_v(s) = 10^4/(1 + 10^{-3}s)$. This op-amp is used in the circuit shown in the figure. The 3-dB bandwidth of the circuit, in rad/s, is

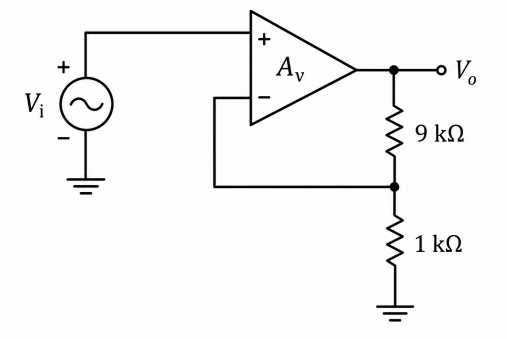


Fig. Q3.15.2

A. 10^2 **B.** 10^3 **C.** 10^4 **D.** 10^6

Q3.15.3 MCQ 2015 1M

The filter whose transfer function is of the form

$$G(s) = \frac{s^2 - bs + c}{s^2 + bs + c}, \; is$$

A. a high-pass filter **B.** a low-pass filter

C. an all-pass filter **D.** a band-reject filter

Q3.14.1 NAT 2014 2M

Assuming an ideal op-amp in linear range of operation, the magnitude of the transfer impedance $\frac{V_o}{i}$ in MΩ of the current to voltage converter shown in the figure is _____ .

Q3.14.2 MCQ 2014 1M

The amplifier in the figure has gain of −10 and input resistance of 50 kΩ. The values of R_i and R_f are

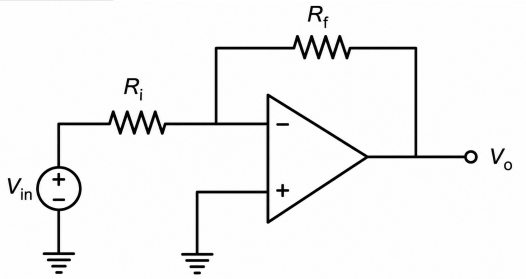


Fig. Q3.14.2

A. R_i = 500 kΩ, R_f = 50 kΩ

B. R_i = 50 kΩ, R_f = 500 kΩ

C. R_i = 5 kΩ, R_f = 10 kΩ

D. R_i = 50 kΩ, R_f = 200 kΩ

Q3.14.3 MCQ 2014 1M

For the op-amp shown in the figure, the bias currents are I_{b1} = 450 nA and I_{b2} = 350 nA. The values of the input bias current (I_B) and the input offset current (I_f) are

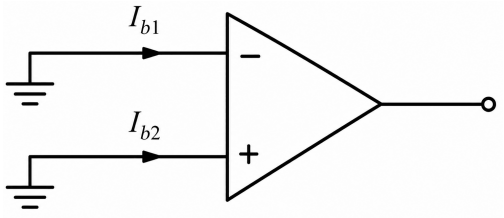


Fig. Q3.14.3

A. I_B = 800 nA, I_f = 50 nA

B. $I_B = 800$ nA, $I_f = 100$ nA

C. $I_B = 400$ nA, $I_f = 50$ nA

D. $I_B = 400$ nA, $I_f = 100$ nA

Q3.14.4 NAT 2014 2M

The figures show an oscillator circuit having an ideal Schmitt trigger and its input-output characteristics. The time period (in ms) of $v_0(t)$ is _____.

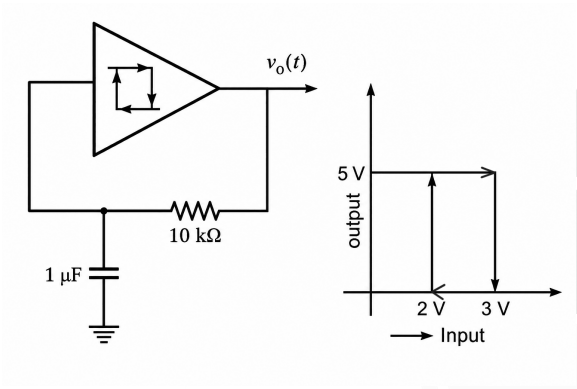


Fig. Q3.14.4

Q3.14.5 NAT 2014 2M

For the given low-pass circuit shown in the figure, the cutoff frequency in Hz will be _____.

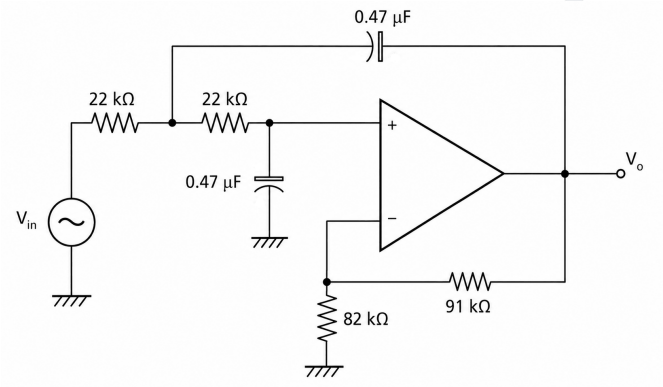


Fig. Q3.14.5

Q3.13.1 MCQ 2013 2M

In the circuit shown below the op-amps are ideal. The V_{out} in volts is

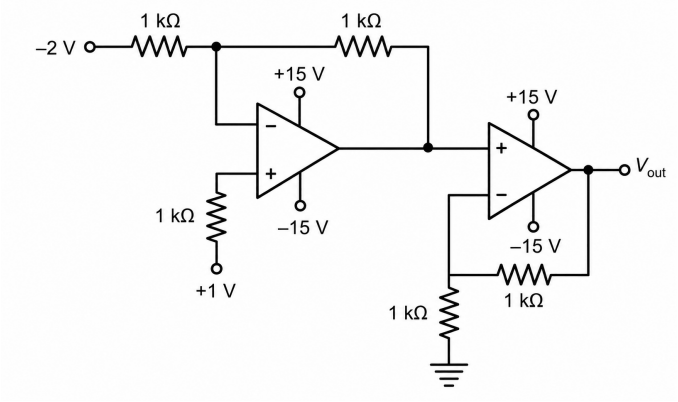


Fig. Q3.13.1

A. 4 **B.** 6 **C.** 8 **D.** 10

Q3.13.2 MCQ 2013 1M

In the circuit shown below what is the output voltage (V_{out}) in volts if a silicon transistor Q and an ideal op-amp are used?

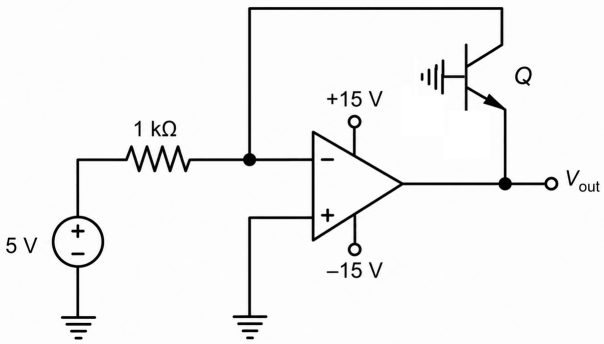


Fig. Q3.13.2

A. −15 **B.** −0.7 **C.** +0.7 **D.** +15

Q3.13.3 MCQ 2013 1M

The operational amplifier shown in the circuit below has a slew rate of 0.8 Volts/μs. The input signal is 0.25 sin(ωt). The maximum frequency of input in kHz for which there is no distortion in the output is

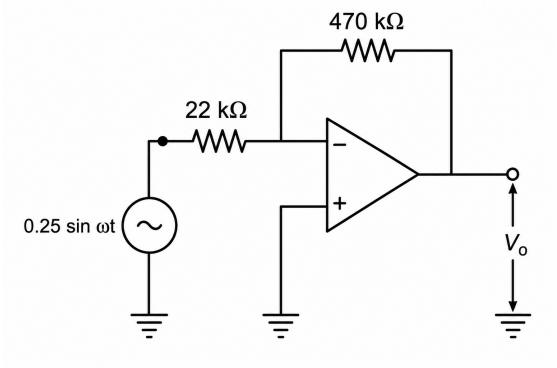


Fig. Q3.13.3

A. 23.84 **B.** 25.0 **C.** 50.0 **D.** 46.60

Q3.13.4 MCQ 2013 2M

A signal $V_i(t) = 10+10\sin 100\pi t+10\sin 4000\pi t+10\sin 100000\pi t$ is supplied to a filter circuit (shown below) made up of ideal op-amps. The least attenuated frequency component in the output will be

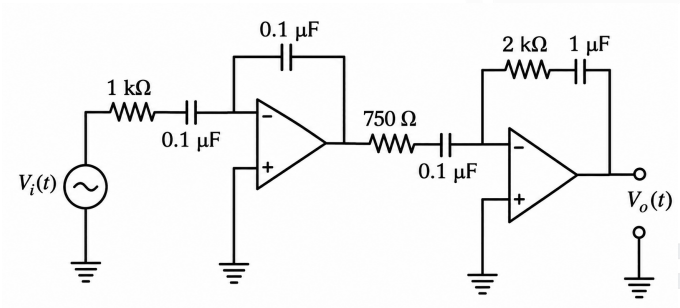


Fig. Q3.13.4

A. 0 Hz **B.** 50 Hz **C.** 2 kHz **D.** 50 kHz

Q3.12.1 MCQ 2012 2M

The circuit shown is a

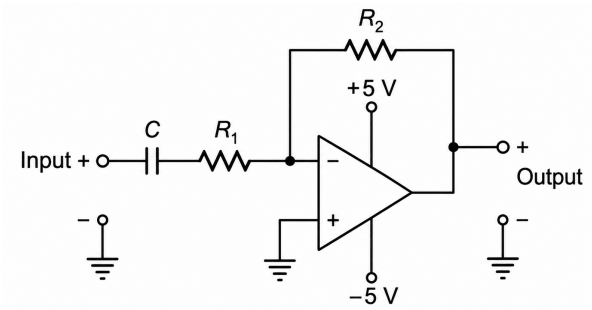


Fig. Q3.12.1

A. low pass filter with $f_{3dB} = \dfrac{1}{(R_1 + R_2)C}$ rad/s

B. high pass filter with $f_{3dB} = \dfrac{1}{R_1 C}$ rad/s

C. low pass filter with $f_{3dB} = \dfrac{1}{R_1 C}$ rad/s

D. high pass filter with $f_{3dB} = \dfrac{1}{(R_1 + R_2)C}$ rad/s

Q3.11.1 MCQ 2011 2M

The value of V_0 of the series regulator shown below is

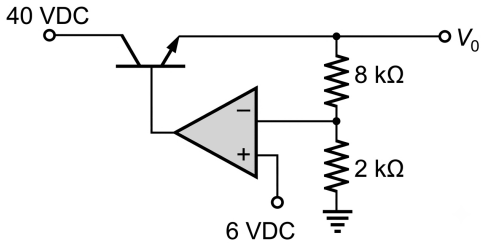


Fig. Q3.11.1

A. 24 V **B.** 28 V **C.** 30 V **D.** 32 V

Q3.11.2 MCQ 2011 2M

Assuming base-emitter voltage of 0.7 V and $\beta = 99$ of transistor Q_1, the output voltage V_0 in the ideal opamp circuit shown below is

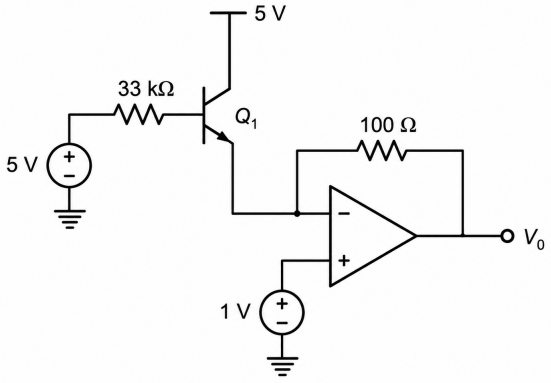


Fig. Q3.11.2

A. −1 V **B.** −1/3.3 V **C.** 0 V **D.** 2 V

Q3.11.3 MCQ 2011 2M

The ideal opamp based circuit shown below acts as a

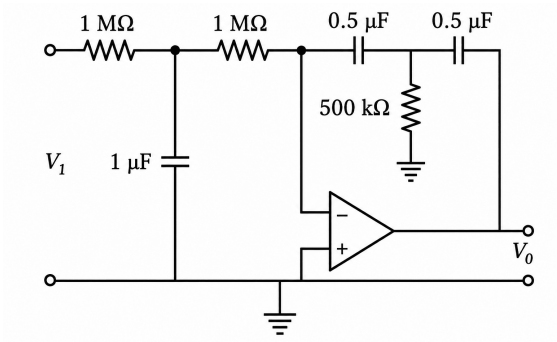


Fig. Q3.11.3

A. low-pass filter **B.** high-pass filter

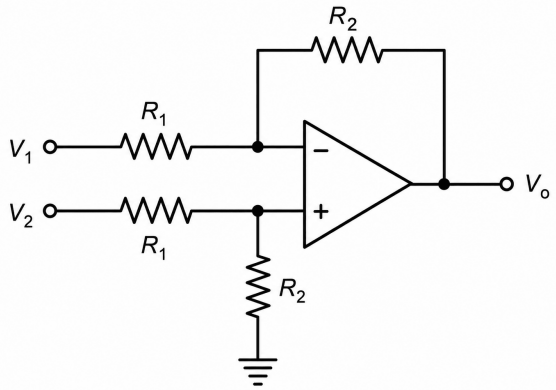
C. band-pass filter **D.** band-reject filter

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q3.10.1 and Q3.10.2

A differential amplifier is constructed using an ideal op-amp as shown in the adjoining figure. The values of R_1 and R_2 are 47 kΩ and 470 kΩ respectively.



Q3.10.1 MCQ 2010 2M

V_1 and V_2 are connected to voltage sources having an open circuit output of +1 V each and internal resistances of 13 kΩ and 3 kΩ respectively. The output voltage V_0 is

A. 0 V **B.** 0.15 V **C.** 1.5 V **D.** 10 V

Q3.10.2 MCQ 2010 2M

The input impedances seen looking into the terminals V_1 and V_2, with respect to ground, respectively are

A. 47 kΩ and 43 kΩ **B.** 47 kΩ and 47 kΩ

C. 47 kΩ and 517 kΩ **D.** 517 kΩ and 517 kΩ

Q3.10.3 MCQ 2010 1M

In the ideal op-amp circuit given in the adjoining figure, the value of R_f is varied from 1 kΩ to 100 kΩ. The gain $G = (V_0/V_i)$

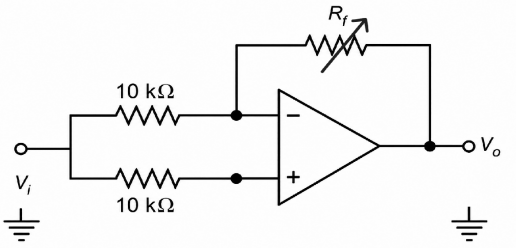


Fig. Q3.10.3

A. remain constant at +1

B. remain constant at −1

C. vary as $-R_f/10,000$

D. vary as $(1 + R_f/10,000)$

Q3.10.4 MCQ 2010 2M

An active filter is shown in the adjoining figure. The dc gain and the 3 dB cut-off frequency of the filter respectively, are nearly

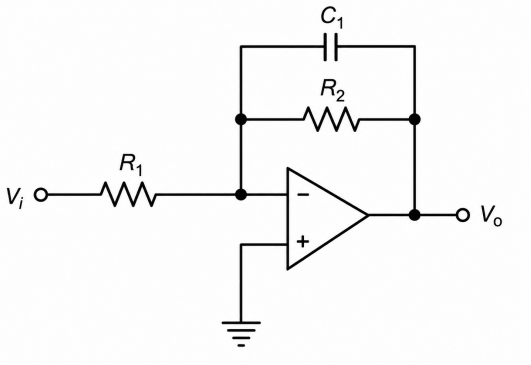


Fig. Q3.10.4

$R_1 = 15.9$ kΩ, $R_2 = 159$ kΩ, $C_1 = 1.0$ nF

A. 40 dB, 3.14 kHz **B.** 40 dB, 1.00 kHz

C. 20 dB, 6.28 kHz **D.** 20 dB, 1.00 kHz

Q3.09.1 MCQ 2009 2M

The input resistance of the circuit shown in the figure, assuming an ideal op-amp is

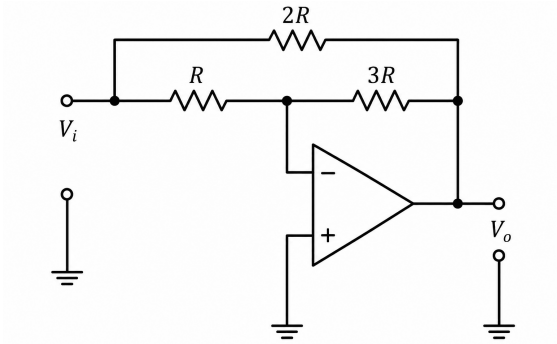


Fig. Q3.09.1

A. $\frac{R}{3}$ **B.** $\frac{2R}{3}$ **C.** R **D.** $\frac{4R}{3}$

Q3.09.2 MCQ 2009 2M

In the circuit shown, the Zener diode has ideal characteristics and a breakdown voltage of 3.2 V. The output voltage V_0 for an input voltage $V_i = +1$ V is closest to

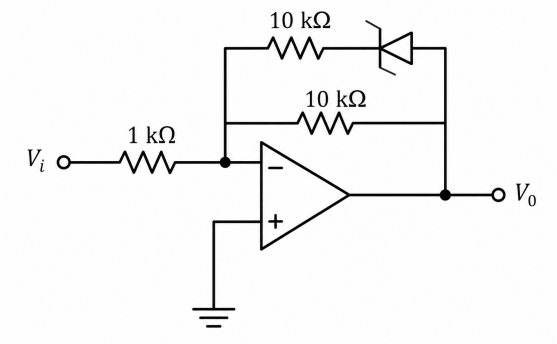


Fig. Q3.09.2

A. −10 V **B.** −6.6 V **C.** −5 V **D.** −3.2 V

Q3.09.3 MCQ 2009 2M

In the circuit shown in the figure, the switch S has been in Position 1 for a long time. It is then moved to Position 2. Assume the Zener diodes to be ideal. The time delay between the switch moving to Position 2 and the transition in the output voltage V_0 is

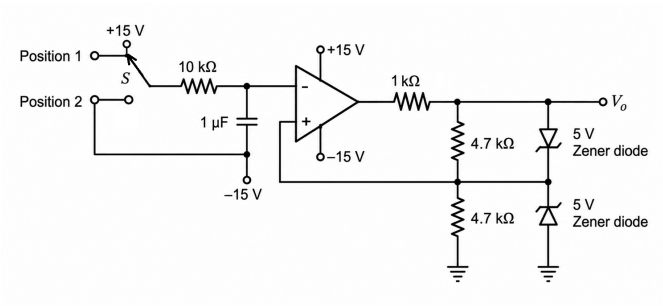


Fig. Q3.09.3

A. 5.00 ms **B.** 8.75 ms **C.** 10.00 ms **D.** 13.75 ms

Q3.09.4 MCQ 2009 1M

The circuit shown in the figure is

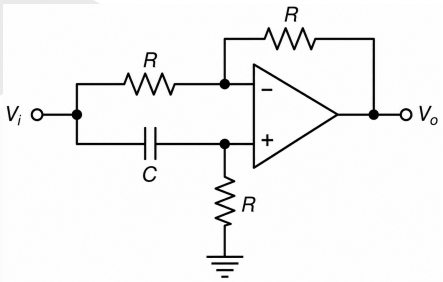


Fig. Q3.09.4

A. an all-pass filter **B.** a band-pass filter

C. a high-pass filter **D.** a low-pass filter

Q3.08.1 MCQ 2008 2M

For the op-amp circuit shown below, v_0 is approximately equal to

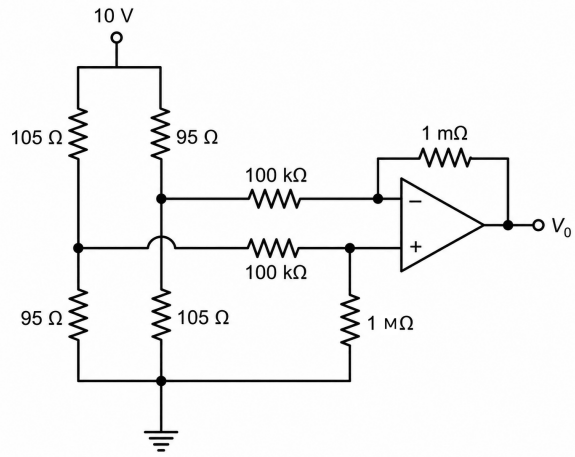


Fig. Q3.08.1

The feedback resistor is 1 MΩ.

A. −10 V **B.** −5 V **C.** +5 V **D.** +10 V

Q3.08.2 MCQ 2008 1M

An ideal op-amp has the characteristics of an ideal

A. voltage controlled voltage source

B. voltage controlled current source

C. current controlled voltage source

D. current controlled current source

Q3.08.3 MCQ 2008 2M

In the op-amp circuit shown below the input voltage V_{in} is gradually increased from −10 V to +10 V. Assuming that the output voltage V_{out} saturates at −10 V and +10 V, V_{out} will change from

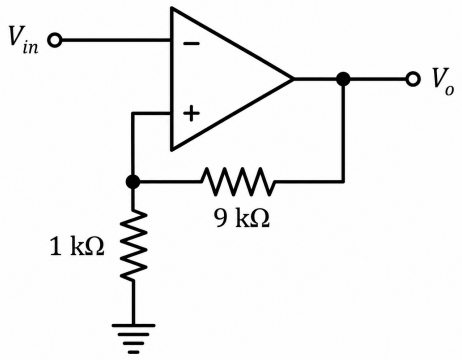


Fig. Q3.08.3

A. −10 V to +10 V when V_{in} = −1 V

B. −10 V to +10 V when V_{in} = +1 V

C. +10 V to −10 V when V_{in} = −1 V

D. +10 V to −10 V when V_{in} = +1 V

Q3.08.4 MCQ 2008 2M

The op-amp circuit shown below is that of a

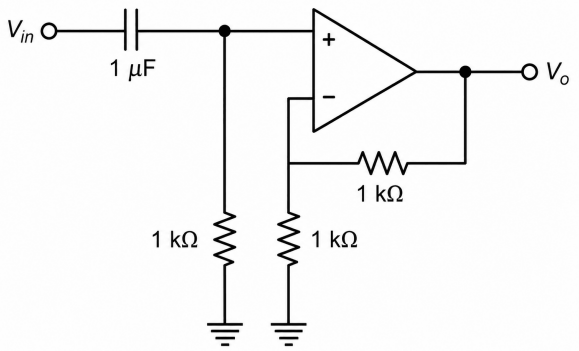


Fig. Q3.08.4

A. low-pass filter with a maximum gain of 1

B. low-pass filter with a maximum gain of 2

C. high-pass filter with a maximum gain of 1

D. high-pass filter with a maximum gain of 2

Q3.08.5 MCQ 2008 2M

A differential amplifier shown below has a differential mode gain of 100 and a CMRR of 40 dB. If V_1 = 0.55 V and V_2 = 0.45 V, output is

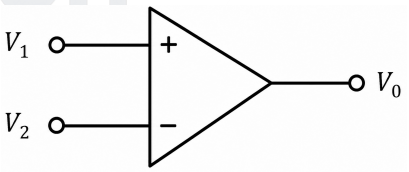


Fig. Q3.08.5

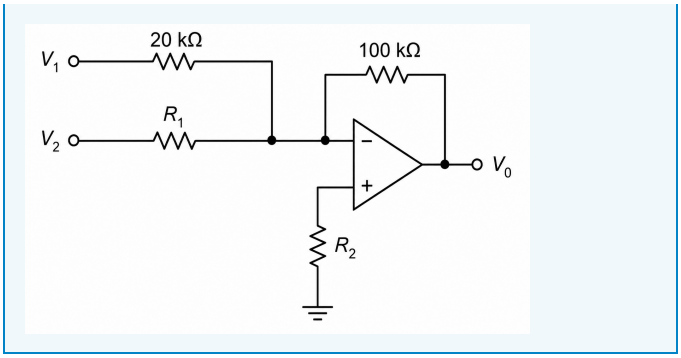
A. 10 V **B.** 10.5 V **C.** 11 V **D.** 15 V

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q3.07.1 and Q3.07.2

Consider the op-amp circuit shown in the figure below



Q3.07.1 MCQ 2007 2M

Let $V_1 = V_2 = 0$ and $R_1 = 20$ kΩ. Assume that the op-amp is ideal except for a non-zero input bias current. What is the value of R_2 for the output voltage of the op-amp to be zero?

A. 2.2 kΩ **B.** 9.1 kΩ **C.** 20 kΩ **D.** 100 kΩ

Q3.07.2 MCQ 2007 2M

If $V_1 = 0.2$ V, $V_2 = 0.6$ V and $V_0 = -7$ V and the op-amp is ideal, the value of R_1 is

A. 5 kΩ **B.** 10 kΩ **C.** 15 kΩ **D.** 20 kΩ

Q3.07.3 MCQ 2007 2M

Let $V_1 = V_2 = V_c \sin 2\pi ft$ and $R_1 = 20$ kΩ. The op-amp has a slew rate of 0.5 V/μs with its other parameters being ideal. The values of V_c and f for which the amplifier output will have no distortion are, respectively

A. 0.1 V and 300 kHz **B.** 0.5 V and 300 kHz
C. 0.1 V and 300 kHz **D.** 0.5 V and 30 kHz

Q3.07.4 MCQ 2007 2M

The figure shows a single op-amp differential amplifier circuit.

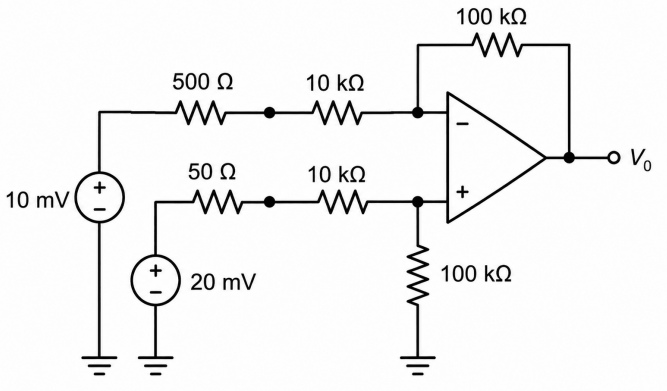


Fig. Q3.07.4

Which one of the following statements about the output is correct?

A. $V_0 \le 95$ mV **B.** 95 mV $< V_0 \le 98$ mV
C. 98 mV $< V_0 \le 101$ mV **D.** $V_0 > 101$ mV

Q3.07.5 MCQ 2007 2M

Consider the linear circuit with an ideal op-amp shown in the figure shown below:

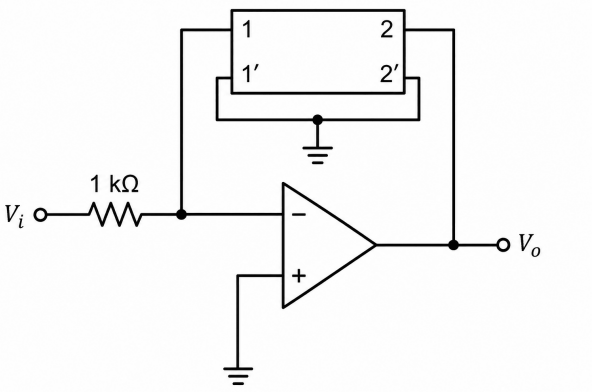


Fig. Q3.07.5

The Z-parameters of two port feedback network are:

$$Z_{11} = Z_{22} = 11 \text{ k}\Omega, \quad \text{and} \quad Z_{12} = Z_{21} = 1 \text{ k}\Omega.$$

The gain of the amplifier is

A. +110 **B.** +11 **C.** −1 **D.** −120

Q3.07.6 MCQ 2007 2M

Consider the triangular wave generator shown below.

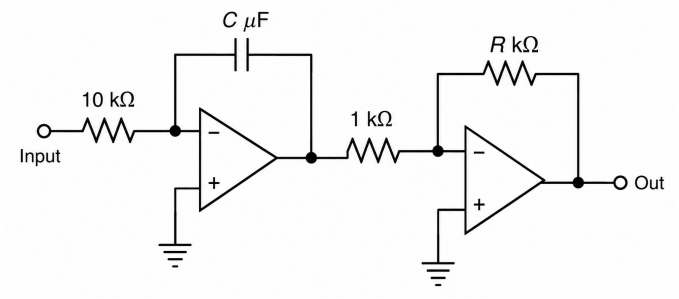


Fig. Q3.07.6

Assume that the op-amps are ideal and have ±12 V power supply. If the input is a ±5 V 50 Hz square wave of duty cycle 50%, the condition that results in a triangular wave of peak to peak amplitude 5 V and frequency 50 Hz at the output is

A. $RC = 1$ **B.** $\frac{R}{C} = 1$

C. $\frac{R}{C} = 5$ **D.** $\frac{C}{R} = 5$

Q3.07.7 MCQ 2007 2M

In the circuit shown below the switch (S) is closed whenever the input voltage (V_{in}) is positive and open otherwise.

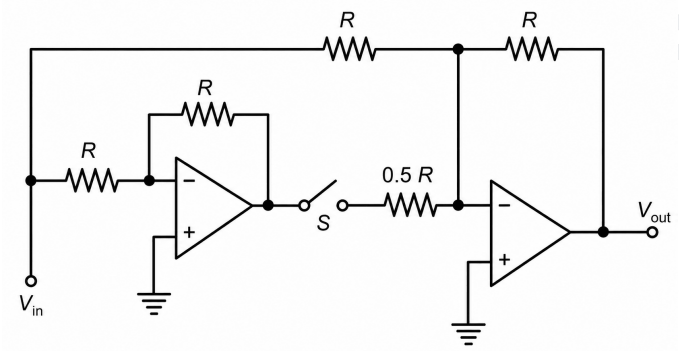


Fig. Q3.07.7

The circuit is a

A. low-pass filter **B.** level shifter

C. modulator **D.** precision rectifier

Q3.07.8 MCQ 2007 2M

Consider the circuit shown below:

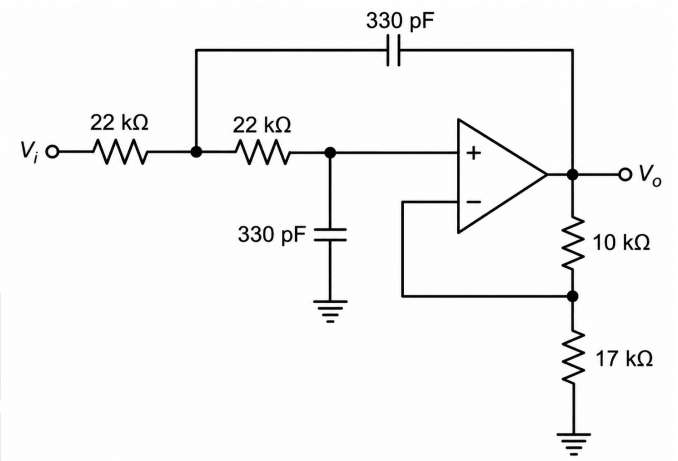
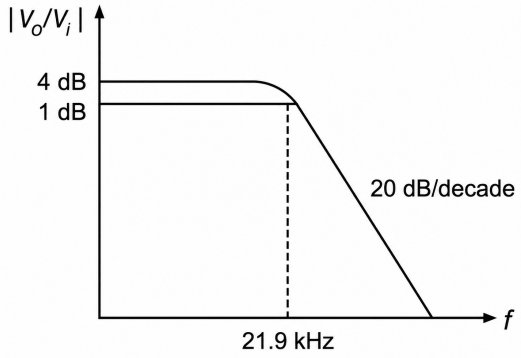


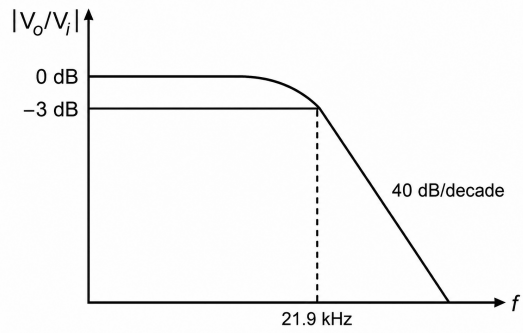
Fig. Q3.07.8

The correct frequency response of the circuit is

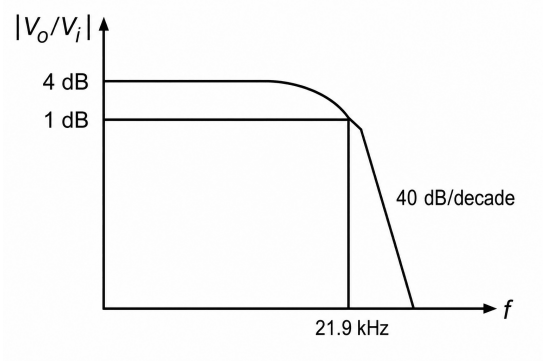
A.



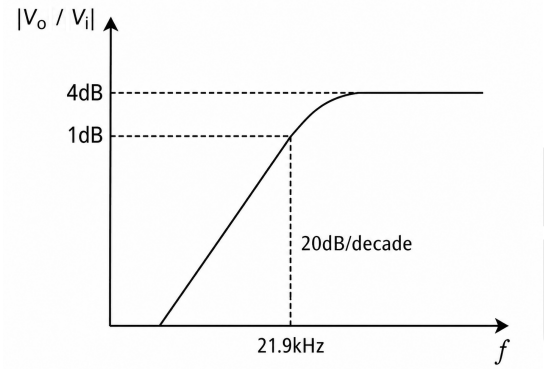
B.



C.



D.



Q3.07.9 MCQ 2007 2M

The input signal shown in the figure shown above is fed to a Schmitt trigger. The signal has a square wave amplitude of 6 V p-p. It is corrupted by an additive high frequency noise of amplitude 8 V p-p.

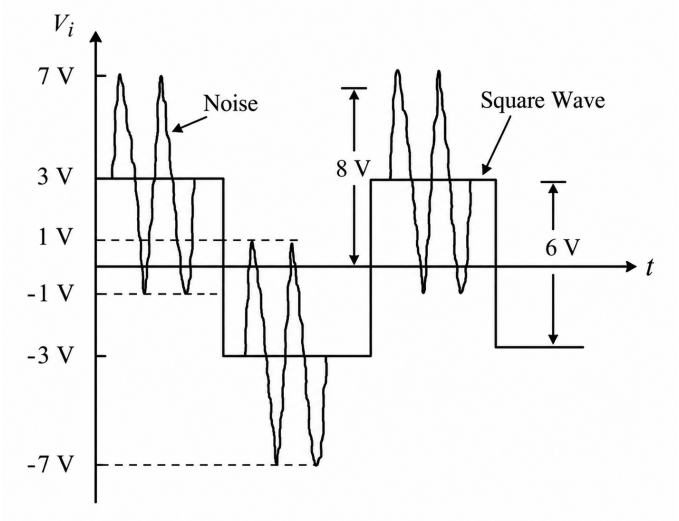


Fig. Q3.07.9

Which one of the following is an appropriate choice for the upper and lower trip points of the Schmitt trigger to recover a square wave of the same frequency from the corrupted input signal V_i ?

A. ±8.0 V **B.** ±2.0 V **C.** ±0.5 V **D.** 0 V

Q3.07.10 MCQ 2007 1M

When light falls on the photodiode shown in the following circuit, the reverse saturation current of the photodiode changes from 100 μA to 200 μA.

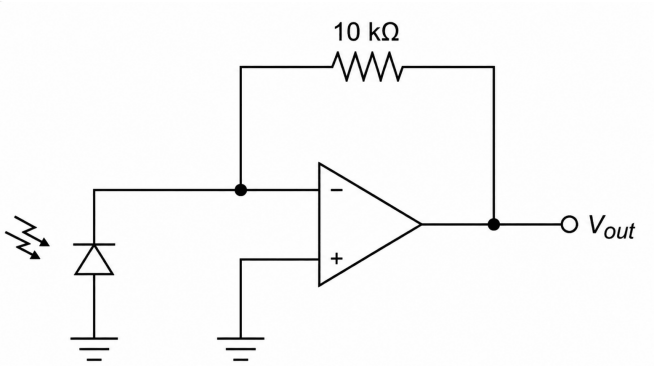


Fig. Q3.07.10

Assuming the op-amp to be ideal, the output voltage, V_{out}, of the circuit

A. does not change

B. changes from 1 V to 2 V

C. changes from 2 V to 1 V

D. changes from −1 V to −2 V

Q3.06.1 MCQ 2006 2M

A dual opamp instrumentation amplifier is shown below. The expression for the output of the amplifier is given by

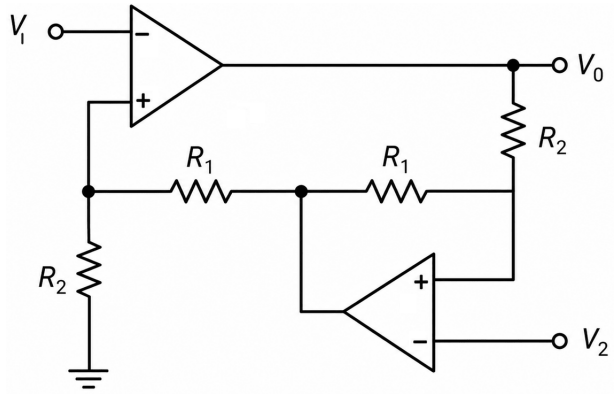


Fig. Q3.06.1

A. $v_0 = \left(1 + \frac{R_2}{R_1}\right)(v_2 - v_1)$ **B.** $v_0 = \left(1 + \frac{2R_2}{R_1}\right)(v_2 - v_1)$

C. $v_0 = \frac{2R_2}{R_1}(v_2 - v_1)$ **D.** $v_0 = \left(1 + \frac{2R_1}{R_2}\right)(v_2 - v_1)$

Q3.06.2 MCQ 2006 2M

The potential difference between the input terminals of an opamp may be treated to be nearly zero, if

A. the two supply voltages are balanced

B. the output voltage is not saturated

C. the opamp is used in a circuit having negative feedback

D. there is a dc bias path between each of the terminals and the circuit ground

Q3.06.3 MCQ 2006 2M

In the circuit shown in the following figure, the opamp has input bias current $I_b < 10$ nA, and input offset voltage $V_{i0} < 1$ mV. The maximum dc error in the output voltage is

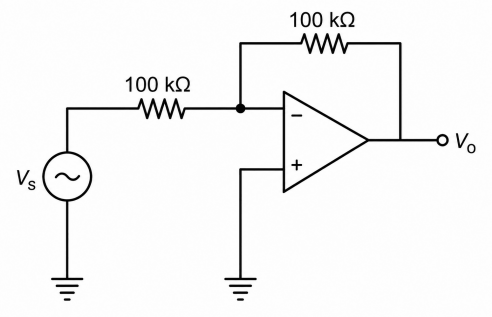


Fig. Q3.06.3

A. 1.0 mV **B.** 2.0 mV **C.** 2.5 mV **D.** 3.0 mV

Q3.06.4 MCQ 2006 1M

When the switch S_2 is closed the gain of the programmable gain amplifier shown in the following figure is

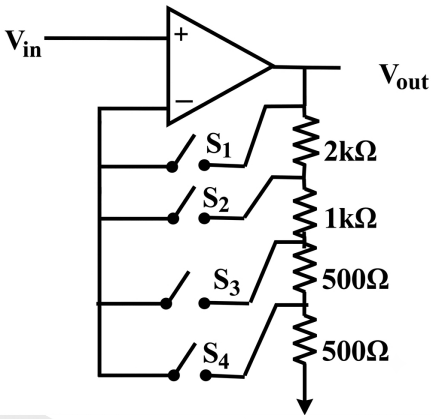


Fig. Q3.06.4

A. 0.5 **B.** 2 **C.** 4 **D.** 8

Q3.06.5 MCQ 2006 1M

If the value of the resistance R in the following figure is increased by 50%, then voltage gain of the amplifier shown in the figure will change by

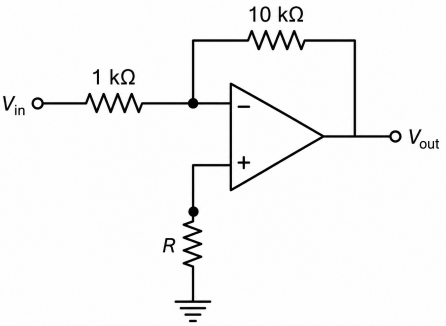


Fig. Q3.06.5

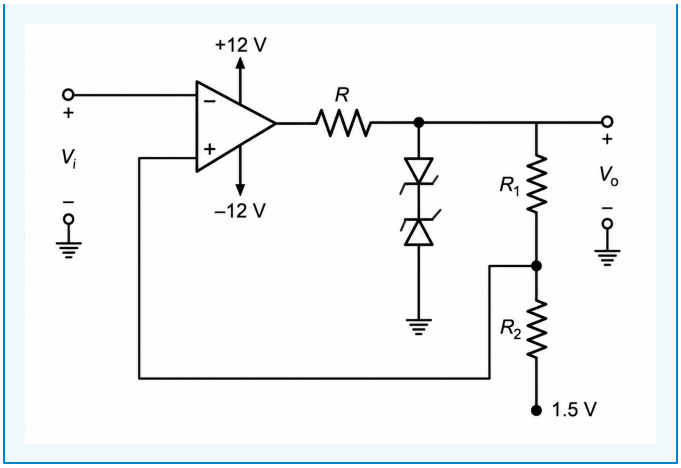
A. 50% **B.** 5%

C. −50% **D.** negligible amount

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q3.06.6 and Q3.6.7 In the Schmitt trigger circuit shown below, the Zener diodes have V_Z (reverse saturation voltage) = 6 V and V_D (forward voltage drop) = 0.7 V.



Q3.06.6 MCQ 2006 2M

If the circuit has the input lower trip point (LTP) = 0 V, then value of R_1/R_2 is given as

A. 0.223 **B.** 2.67 **C.** 4.67 **D.** ∞

Q3.06.7 MCQ 2006 2M

The input upper trip point (UTP) of the Schmitt trigger is

A. 1.5 V **B.** 2.1 V **C.** 2.42 V **D.** 6.7 V

Q3.05.1 MCQ 2005 2M

In the instrumentation amplifier shown in the figure if the switch SW is changed from position A to B, the values of the amplifier gain G before and after changing the switch respectively are

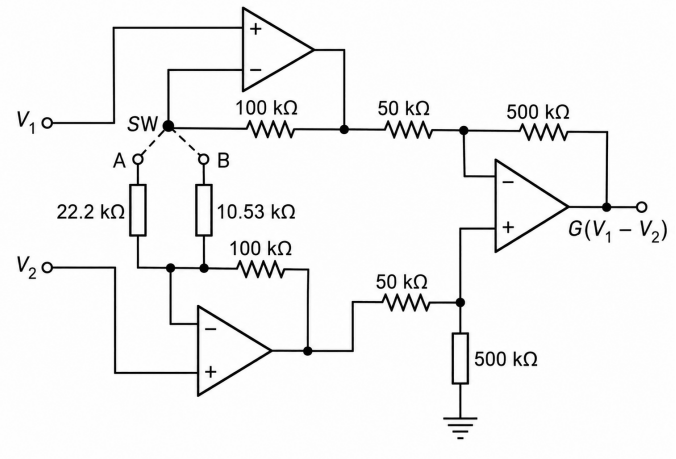


Fig. Q3.05.1

A. 45, 95 **B.** 50, 100 **C.** 100, 200 **D.** 90, 180

Q3.05.2 MCQ 2005 2M

In the circuit shown in the figure, input voltage V_i is symmetrical saw-tooth wave of average value zero, positive slope and peak-to-peak value 20 V. The average value of output, assuming an ideal operational amplifier with peak-to-peak symmetrical swing of 30 V, is

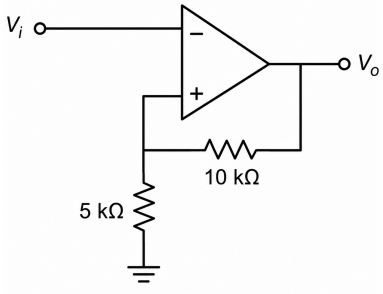


Fig. Q3.05.2

A. 5 V **B.** 10 V **C.** −5 V **D.** 7.5 V

Q3.05.3 MCQ 2005 1M

In the circuit shown in the given figure the input voltage $V_{in}(t)$ is given by $2\sin(100\pi t)$. For R_L in the range 0.5 kΩ to 1.5 kΩ, the current through R_L is

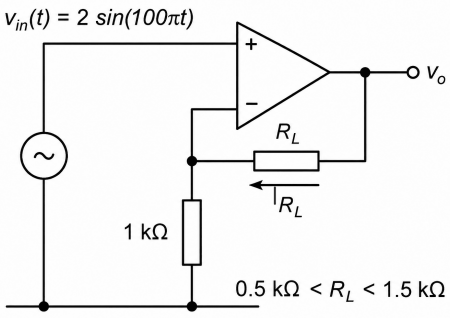


Fig. Q3.05.3

A. $+2\sin(100\pi t)$ mA **B.** $-2\sin(100\pi t)$ mA

C. $+0.5\sin(100\pi t)$ mA **D.** $1.5\sin(100\pi t)$ mA

Q3.05.4 MCQ 2005 2M

The input-output characteristic of a Schmitt trigger has a hysteresis band of ±0.1 V. If the input voltage is $5\sin(1000\pi t)$, the delay between the corresponding zero cross-over points of the output and input signals is

A. 6.37 μs **B.** 0.02 μs **C.** 63.7 μs **D.** 2.0 μs

Q3.04.1 MCQ 2004 2M

V_1 and V_2 are the input voltages of an instrumentation amplifier. The output of the instrumentation amplifier is found to be $100(V_1 - V_2) + 10^{-4}(V_1 + V_2)$. The gain and the common mode rejection ratio (CMRR) of the instrumentation amplifier respectively are

A. (50, 60 dB) **B.** (50, 120 dB)

C. (100, 60 dB) **D.** (100, 120 dB)

Q3.04.2 MCQ 2004 2M

The gain $\left(\frac{V_0}{V_i}\right)$ of the amplifier circuit shown in figure is

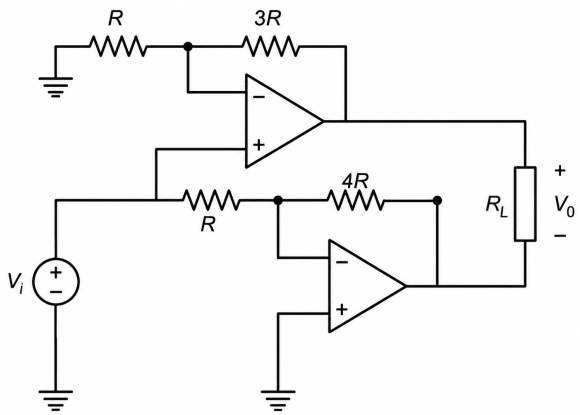


Fig. Q3.04.2

A. 8 **B.** 4 **C.** −4 **D.** $\frac{3R_L}{R}$

Q3.04.3 MCQ 2004 2M

The output voltage V_0 in the circuit in figure is

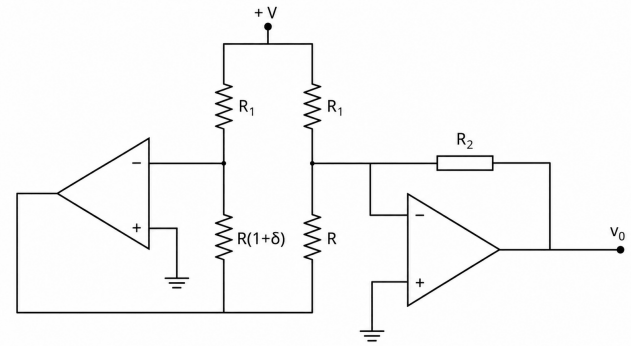


Fig. Q3.04.3

A. $\frac{R_2}{R}V\delta$ **B.** $\frac{R_2}{R_1}V\delta$

C. $\frac{R_2}{R_1\delta}V$ **D.** $\frac{R_2}{R(1+\delta)}V$

Q3.04.4 MCQ 2004 2M

The value of V_0 in the circuit, shown in figure is

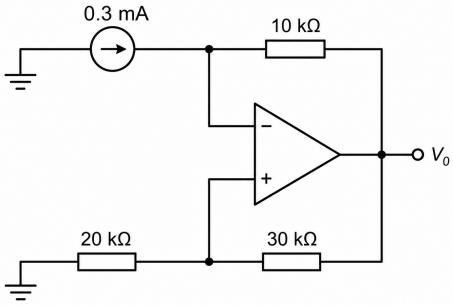


Fig. Q3.04.4

A. −5 V **B.** −3 V **C.** +3 V **D.** +5 V

Q3.04.5 MCQ 2004 1M

The output of the opamp in the circuit of figure is

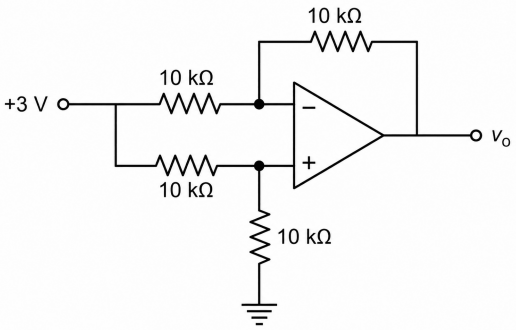


Fig. Q3.04.5

A. 0 V **B.** −3 V **C.** +1.5 V **D.** +3 V

Q3.04.6 MCQ 2004 2M

Figure (a) shows a Schmitt trigger circuit and figure (b) the corresponding hysteresis characteristics. The values of V_{TL} and V_{TH} are

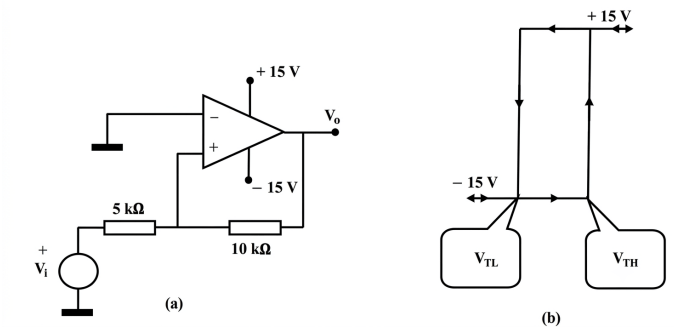


Fig. Q3.04.6

A. $V_{TL} = -3.75$ V, $V_{TH} = +3.75$ V

B. $V_{TL} = -1$ V, $V_{TH} = +5$ V

C. $V_{TL} = -5$ V, $V_{TH} = +1$ V

D. $V_{TL} = -5$ V, $V_{TH} = +5$ V

Q3.04.7 MCQ 2004 2M

The circuit in figure is a

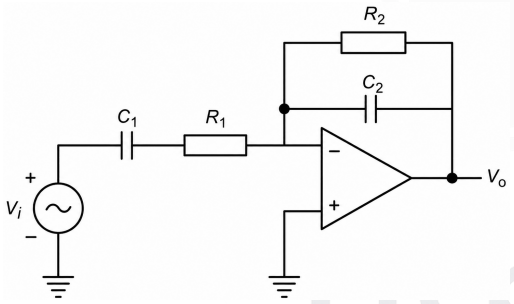


Fig. Q3.04.7

A. band-pass filter with lower cut-off $\omega_L = \frac{1}{R_1C_1}$ and higher cut-off $\omega_H = \frac{1}{R_2C_2}$

B. band-reject filter with lower cut-off $\omega_L = \frac{1}{R_1C_1}$ and higher cut-off $\omega_H = \frac{1}{R_2C_2}$

C. band-pass filter with lower cut-off $\omega_L = \frac{1}{R_2C_2}$ and higher cut-off $\omega_H = \frac{1}{R_1C_1}$

D. band-reject filter with lower cut-off $\omega_L = \frac{1}{R_2C_2}$ and higher cut-off $\omega_H = \frac{1}{R_1C_1}$

Q3.03.1 MCQ 2003 2M

In the circuit shown in figure, the opamps used are ideal. The output V_0 is

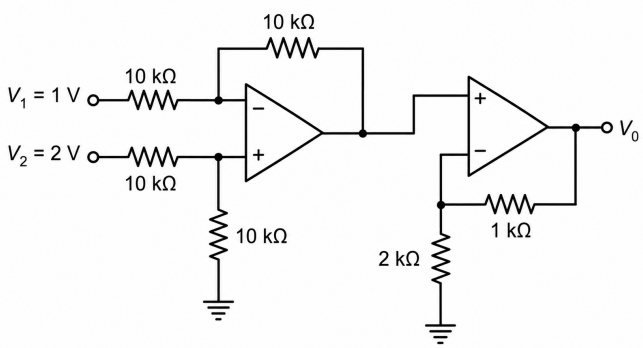


Fig. Q3.03.1

A. 3.0 V **B.** 1.5 V **C.** 1.0 V **D.** 0.5 V

Q3.03.2 MCQ 2003 2M

The opamp used in the inverting amplifier shown in figure has an equivalent input offset voltage V_{ios} of 5 mV. The output offset voltage is

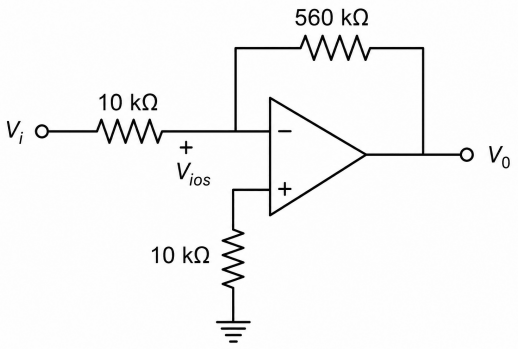


Fig. Q3.03.2

A. 5 mV **B.** 280 mV **C.** 285 mV **D.** 560 mV

Q3.03.3 MCQ 2003 1M

The opamp and the 1 mA current source in the circuit of figure are ideal. The output of the op-amp is

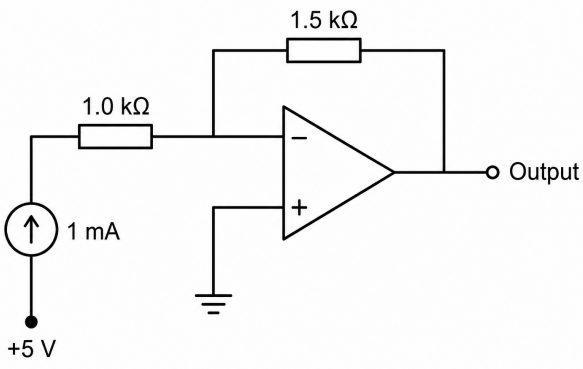


Fig. Q3.03.3

A. −1.5 mA **B.** −1.5 V **C.** −7.5 V **D.** +1.5 V

Q3.03.4 MCQ 2003 1M

The output of an opamp whose input is a 2.5 MHz square wave is shown in figure. The slew rate of the opamp is

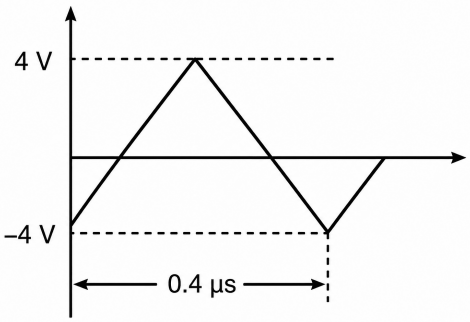


Fig. Q3.03.4

A. 0.8 V/μs **B.** 8.0 V/μs **C.** 20.0 V/μs **D.** 40.0 V/μs

Q3.03.5 MCQ 2003 2M

The circuit shown in figure is that of a waveform generator. Assuming ideal devices, and ±12 V supply, the output V_0 is a

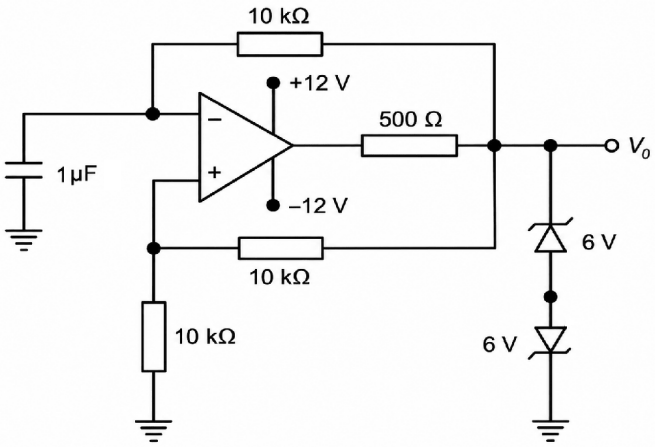


Fig. Q3.03.5

A. triangular wave of period 120 ms and amplitude ±6 V

B. square wave of period 60 ms and amplitude ±6 V

C. square wave of period 120 ms and amplitude ±6 V

D. square wave of period 60 ms and amplitude ±12 V

Q3.03.6 MCQ 2003 1M

The input-output characteristics of a Schmitt trigger is given in below figure. The noise margin of the Schmitt trigger is

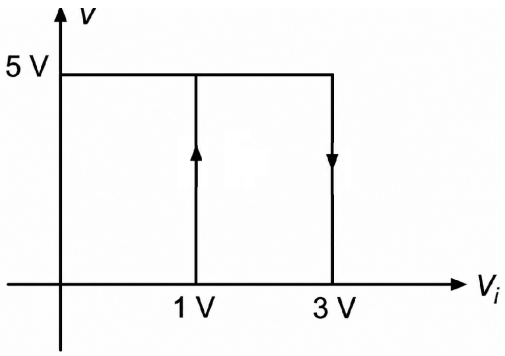


Fig. Q3.03.6

A. 1 V **B.** 2 V **C.** 3 V **D.** 4 V

Q3.03.7 MCQ 2003 1M

An integrator circuit is shown in figure. The op-amp is of type 741 and has an input offset current i_{os} of 1 μA. C is 1 μF and R is 1 MΩ. If the input V_i is a 1 kHz square wave of 1 V peak to peak, the output V_0, under steady condition, will be

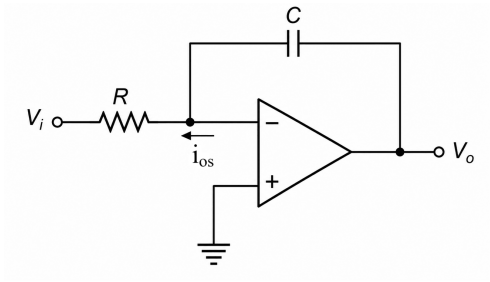


Fig. Q3.03.7

A. a square wave of 1 V peak to peak

B. a triangular wave of 1 V peak to peak

C. positive supply voltage $+V_{cc}$

D. negative supply voltage $-V_{cc}$

Q3.03.8 MCQ 2003 2M

A resistance R_T of a resistive transducer is modeled as $R_T = R(1-Kx)$, where K is the constant of transformation and x the input quantity being sensed. The resistor is connected to an ideal, opamp signal conditioning circuit given in figure. The value of R is 100 Ω, K is 0.004 and x is 75. The output V_0 is

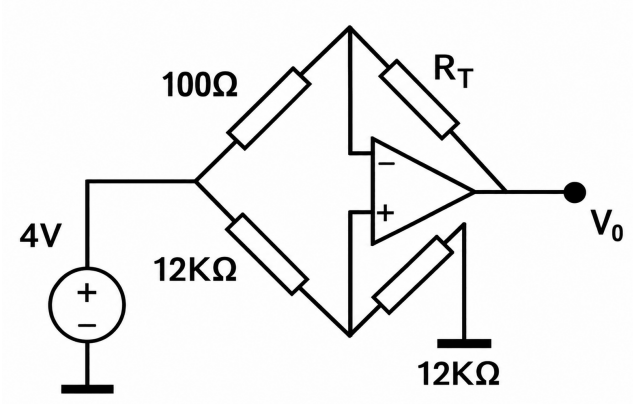


Fig. Q3.03.8

A. −400 mV **B.** −4 mV
C. −300 mV **D.** +600 mV

Q3.03.9 MCQ 2003 2M

A photo-detector circuit is given in the Figure. The photo-diode has an active area of 10^{-2} cm^2 and sensitivity of 0.55 A/W. When light of intensity 10 mW/cm^2 falls on the photo-diode, the output V_0 is

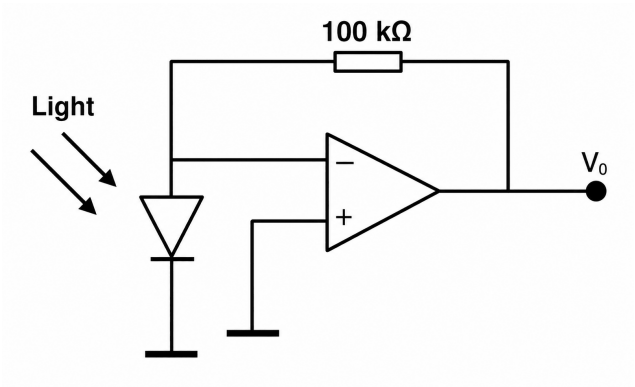


Fig. Q3.03.9

A. 10.1 V **B.** 5.5 V **C.** −5.5 V **D.** −10.1 V

Q3.03.10 MCQ 2003 2M

A Piezoelectric transducer with a sensitivity of 2.0 pC/N having a capacitance of 1600 pF and a leakage resistance of 10^{12} Ω is connected to a charge amplifier as shown in figure. If a force of 0.1 sin 10t N is applied to the transducer, the output-amplitude of the charge amplifier is

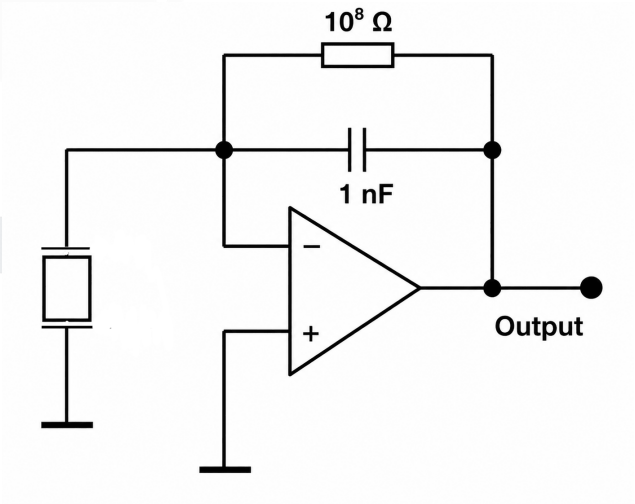


Fig. Q3.03.10

A. 0.141 mV **B.** 0.232 mV
C. 1.414 mV **D.** 1.732 mV

Q3.02.1 MCQ 2002 1M

An amplifier of gain 10 with a gain BW product of 1 MHz and slew rate of 0.1 V/μsec is fed with a 10 kHz symmetrical square wave of ±10 V amplitude. Its output will be

A. ±10 V amplitude square wave

B. ±2.5 V amplitude square wave

C. ±10 V amplitude triangular wave

D. ±2.5 V amplitude triangular wave

Q3.01.1 MCQ 2001 1M

In the figure shown input offset voltage of the operational amplifier is 2 mV. The output DC-error voltage is

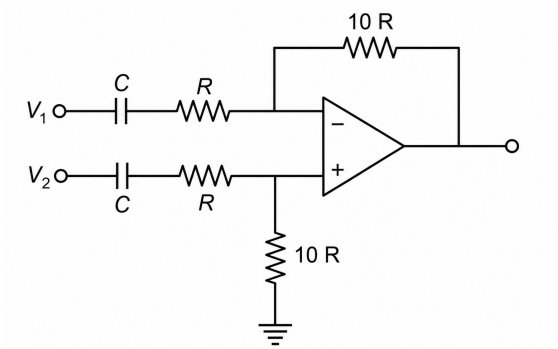


Fig. Q3.01.1

A. 0 **B.** 2 mV **C.** 11 mV **D.** 22 mV

Q3.01.2 MCQ 2001 1M

A sample and hold circuit have two buffers one at input and other at the output. The primary requirements for the buffers are

A. The input buffer should have high slew rate and output buffer should have low bias current.

B. The input buffer should have low bias current and output buffer should have high slew rate.

C. both the buffers should have low bias currents.

D. both the buffers should have high slew rate.

Q3.00.1 MCQ 2000 1M

An op-amp with a slew rate of 1 V/μsec has been used to build an amplifier of gain +10. If the input to the amplifier is a sinusoidal voltage having peak amplitude of 1 V, the maximum allowable frequency of the input signal for undistorted output is

A. 830 Hz **B.** 15.92 kHz **C.** 31.84 kHz **D.** 1 MHz

Q3.99.1 MCQ 1999 1M

The op-amp in the amplifier circuit shown in figure has an offset voltage of 10 mV and it is ideal otherwise. If V_i is zero, the output voltage V_0 is

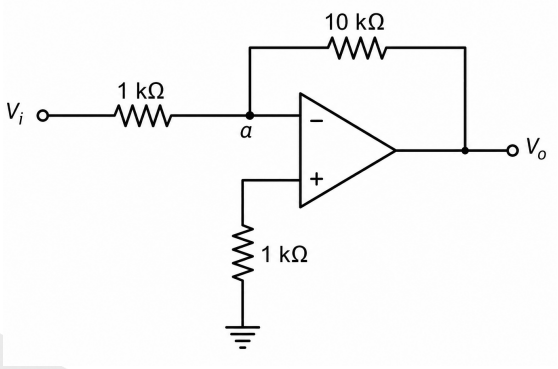


Fig. Q3.99.1

A. 0 **B.** 10 mV **C.** 100 mV **D.** 110 mV

Q3.99.2 MCQ 1999 1M

Pick up the correct relationship for the circuit shown in figure

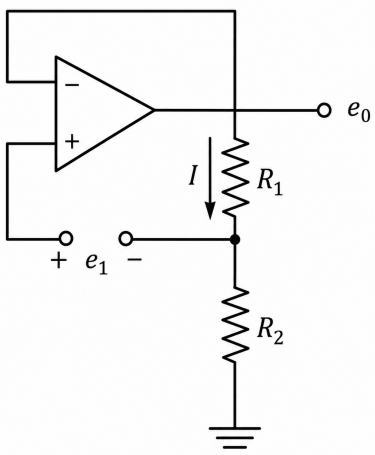


Fig. Q3.99.2

A. $e_0 = e_1$ **B.** $e_0 = \frac{R_2}{R_1} e_1$

C. $I = \frac{e_1}{R_2}$ **D.** $I = \frac{e_1}{R_1}$

Q3.97.1 MCQ 1997 1M

In the op-amp circuit shown in figure the output voltage V_0 is equal to

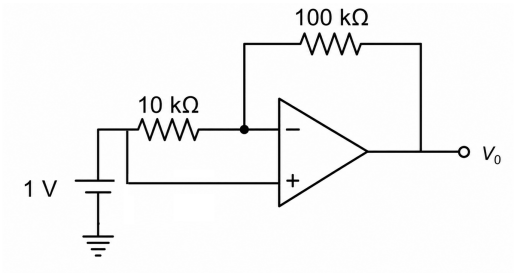


Fig. Q3.97.1

A. 11 V **B.** −10 V **C.** 9 V **D.** 1 V

Q3.97.2 MCQ 1997 1M

The operational amplifier use a differential input stage with a constant current source mainly to obtain

A. Very low common mode gain

B. Very high differential gain

C. Very low input noise

D. Very high input resistance

Q3.96.1 MCQ 1996 1M

The differential input resistance of the circuit shown in figure is

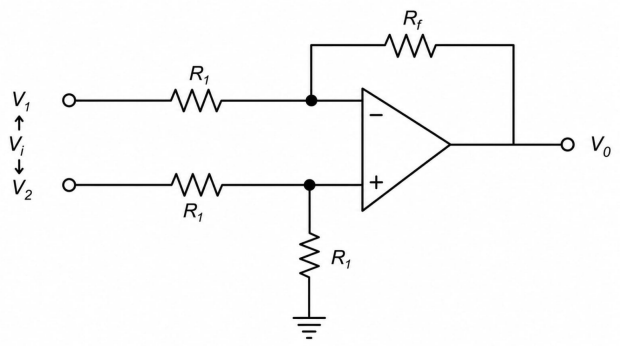


Fig. Q3.96.1

A. R_1 **B.** $R_1/2$ **C.** $2R_1$ **D.** R_f

Q3.96.2 MCQ 1996 1M

In the circuit shown in figure, if $e_i = \sin t$, the voltage V_0 is

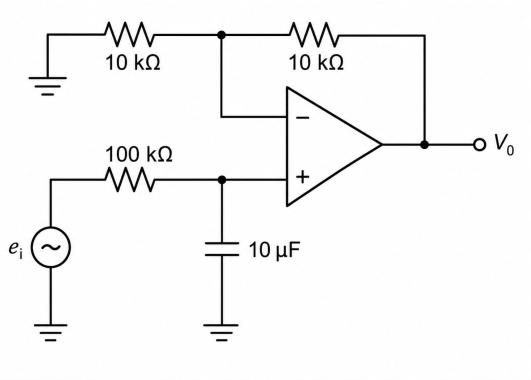


Fig. Q3.96.2

A. $\sqrt{2}\sin\left(t-\frac{\pi}{4}\right)$ **B.** $\sqrt{2}\sin\left(t+\frac{\pi}{4}\right)$

C. $\frac{1}{\sqrt{2}}\sin\left(t-\frac{\pi}{4}\right)$ **D.** $\frac{1}{\sqrt{2}}\sin\left(t+\frac{\pi}{4}\right)$

Q3.95.1 NAT 1995 1M

The Common Mode Rejection Ratio (CMRR) of an op-amp with a common mode gain of 0.01 and a differential mode gain of 10^5 is _____ dB

Q3.94.1 MCQ 1994 1M

The input resistance of the op-amp circuit shown in figure is

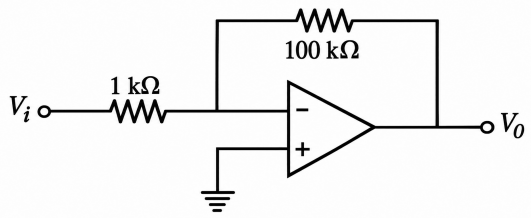


Fig. Q3.94.1

A. 100 kΩ **B.** 99 kΩ **C.** 1 kΩ **D.** 101 kΩ

Q3.93.1 MCQ 1993 1M

The cut-in voltage of diodes in the rectifier of figure shown is 0.6 V. Identify the correct output and input characteristic (V_0 vs V_1)

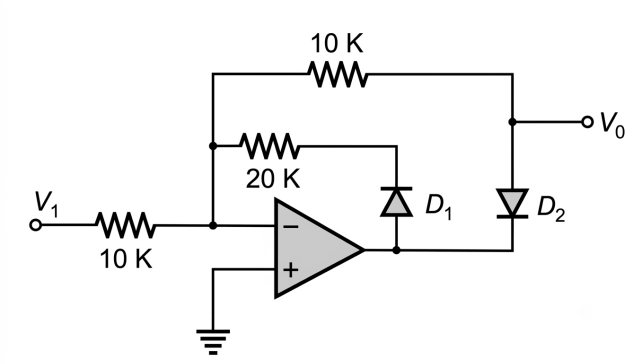
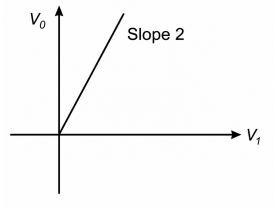
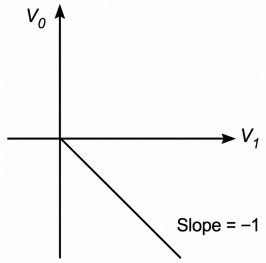
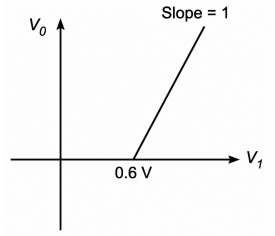
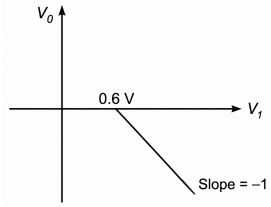


Fig. Q3.93.1

A.


B.


C.


D.


Q3.92.1 NAT 1992 1M

For an ideal operational amplifier in the circuit shown in figure, the transfer function $\frac{V_0(s)}{V_i(s)}$ is

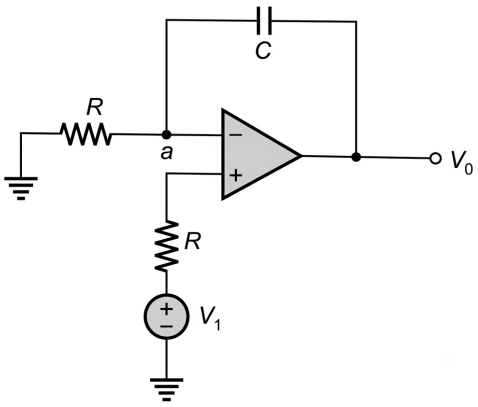


Fig. Q3.92.1

Q3.92.2 MCQ 1992 1M

The operational amplifier is used in the non linear mode in

A. Integrators **B.** Active filters

C. Schmitt triggers **D.** Instrumentation amplifiers

Q3.92.3 MCQ 1992 1M

To realize a phase shift oscillator,

A. It is possible to use a voltage follower as an amplifier along with a suitable phase shift network.

B. Only common emitter or common source stages can be used.

C. Given any phase shifting network, it is always possible to find the emitter follower.

D. None of the above

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q3.26.1 C	Q3.26.2 C	Q3.25.1 A
Q3.25.2 B,C,D	Q3.24.1 1.6	Q3.24.2 -3
Q3.24.3 A	Q3.24.4 A	Q3.24.5 4
Q3.24.6 6.00 to 6.30	Q3.24.7 9.5 to 10.5	Q3.24.8 -2.6 to -2.4
Q3.23.1 1110	Q3.23.2 0.49 to 0.51	Q3.23.3 99 to 101
Q3.23.4 1.9 to 2.1	Q3.22.1 9.9 to 10	Q3.22.2 15.80 to 16
Q3.22.3 0.365 to 0.369	Q3.21.1 1	Q3.21.2 C
Q3.21.3 1.0 to 1.1	Q3.20.1 7	Q3.20.2 -4
Q3.20.3 1	Q3.20.4 0.5	Q3.20.5 3.5
Q3.19.1 9 to 9.1	Q3.19.2 C	Q3.18.1 0.98 to 1.00
Q3.18.2 17.5 to 17.9	Q3.18.3 MTA	Q3.17.1 -5
Q3.17.2 1.4 to 1.6	Q3.17.3 B	Q3.17.4 0.6 to 0.7

Q3.17.5 2.9 to 3.0	Q3.17.6 A	Q3.16.1 100
Q3.16.2 100	Q3.16.3 40 to 41	Q3.16.4 20
Q3.16.5 1.55 to 1.65	Q3.16.6 -1	Q3.15.1 1.3 to 1.4
Q3.15.2 D	Q3.15.3 C	Q3.14.1 0.6
Q3.14.2 B	Q3.14.3 D	Q3.14.4 8.0 to 8.5
Q3.14.5 15 to 16	Q3.13.1 C	Q3.13.2 B
Q3.13.3 A	Q3.13.4 C	Q3.12.1 B
Q3.11.1 C	Q3.11.2 C	Q3.11.3 A
Q3.10.1 B	Q3.10.2 C	Q3.10.3 A
Q3.10.4 D	Q3.09.1 A	Q3.09.2 B
Q3.09.3 A	Q3.09.4 A	Q3.08.1 B
Q3.08.2 A	Q3.08.3 D	Q3.08.4 D
Q3.08.5 B	Q3.07.1 B	Q3.07.2 B
Q3.07.3 C	Q3.07.4 B	Q3.07.5 D
Q3.07.6 B	Q3.07.7 D	Q3.07.8 C
Q3.07.9 B	Q3.07.10 D	Q3.06.1 A
Q3.06.2 C	Q3.06.3 D	Q3.06.4 B
Q3.06.5 D	Q3.06.6 C	Q3.06.7 C
Q3.05.1 C	Q3.05.2 D	Q3.05.3 A
Q3.05.4 A	Q3.04.1 D	Q3.04.2 A
Q3.04.3 B	Q3.04.4 A	Q3.04.5 A
Q3.04.6 MTA	Q3.04.7 A	Q3.03.1 B
Q3.03.2 C	Q3.03.3 B	Q3.03.4 D
Q3.03.5 C	Q3.03.6 B	Q3.03.7 C

Q3.03.8 D	Q3.03.9 C	Q3.03.10 A
Q3.02.1 D	Q3.01.1 B	Q3.01.2 A
Q3.00.1 B	Q3.99.1 D	Q3.99.2 D
Q3.97.1 D	Q3.97.2 A	Q3.96.1 C
Q3.96.2 A	Q3.95.1 140	Q3.94.1 C
Q3.93.1 B	Q3.92.1 *	Q3.92.2 C
Q3.92.3 B		

UNIT

IV

BJTs

Topics

1. BJT Physics and Biasing
2. Current Mirror Circuit Using BJT
3. Small Signal Analysis of BJT Amplifiers
4. Differential Amplifiers and Feedback Topologies Using BJTs

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 28	**MCQ:** 17	**MSQ:** 0	**NAT:** 10
1M: 13	**2M:** 15	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–3M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MCQ 2026 1M

In the circuit shown, the BJT has a β of 100. The base-emitter junction voltage is 0.65 V. The quiescent collector current, I_C is _____ mA.

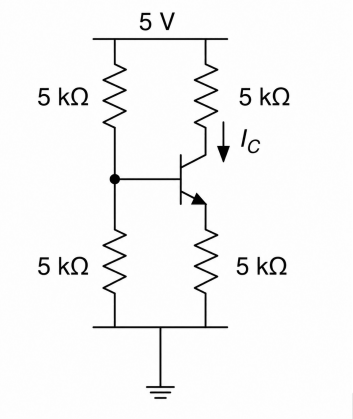


Fig. Q4.26.1

A. 0.365 **B.** 0.435 **C.** 0.625 **D.** 0.862

Q4.25.1 NAT 2025 2M

In the circuit shown, assume that the BJT in the circuit has very high β and $V_{BE} = 0.7$ V, and the Zener diode has $V_Z = 4.7$ V. The current I through the LED is _____ mA (rounded off to two decimal places).

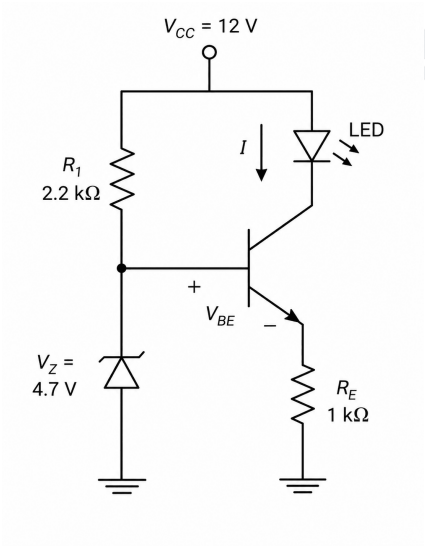


Fig. Q4.25.1

Q4.22.1 NAT 2022 2M

In the circuit shown, $R_1 = 100$ kΩ and $R_2 = 1$ kΩ. If the base-to-emitter voltage of the npn BJT is 0.7 V and the collector-to-emitter voltage is 5.2 V, the β (current gain) of the BJT is _____ (round off to two decimal places).

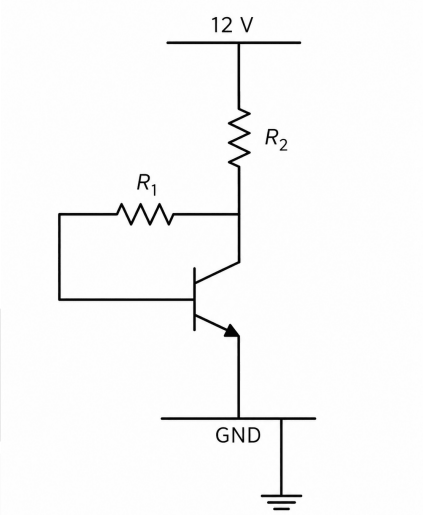


Fig. Q4.22.1

Q4.21.1 NAT 2021 1M

The transistor Q_1 has a current gain $\beta_1 = 99$ and the transistor Q_2 has a current gain $\beta_2 = 49$. The current I_{B2} in microampere is _____.

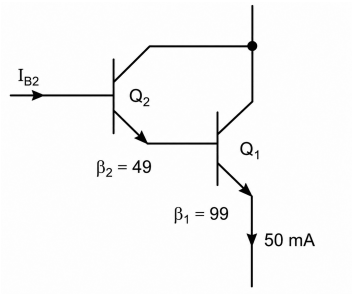


Fig. Q4.21.1

Q4.21.2 NAT 2021 2M

All the transistors used in the circuit are matched and have a current gain β of 20. Neglecting the Early effect, the current I_{O4} in milliampere is _____.

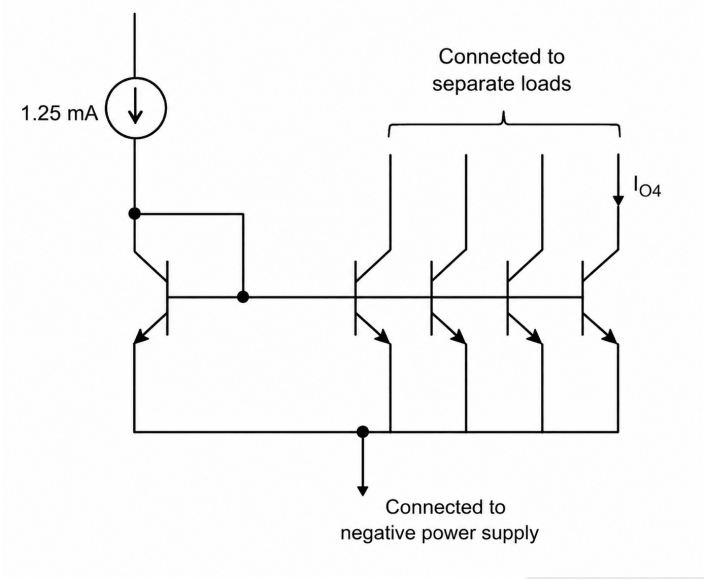


Fig. Q4.21.2

Q4.18.1 NAT 2018 1M

In the given circuit, assume that the opamp is ideal and the transistor has a β of 20. The current I_0 (μA) flowing through the load Z_L is _____ .

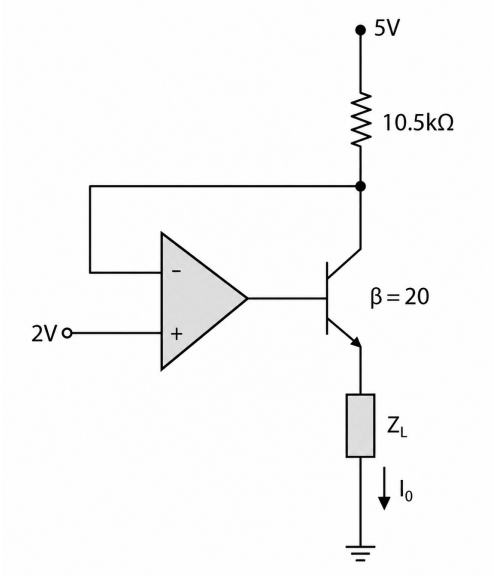


Fig. Q4.18.1

Q4.15.1 NAT 2015 2M

In the circuit shown in the figure, it is found that V_{BE} = 0.7 V and V_E = 0 V.

If β_{dc} = 99 for the transistor, then the value of R_B in kilo ohms is _____ kΩ.

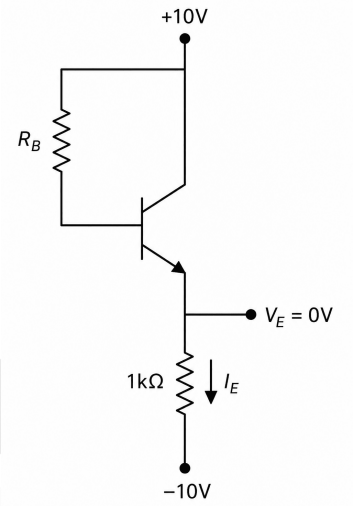


Fig. Q4.15.1

Q4.14.1 NAT 2014 1M

In the figure, transistors T_1 and T_2 have identical characteristics. $V_{CE(\text{sat})}$ of transistor T_3 is 0.1 V. The voltage V_1 is high enough to put T_3 in saturation. Voltage V_{BE} of transistors T_1, T_2 and T_3 is 0.7 V. The value of $(V_1 - V_2)$ in V is _____.

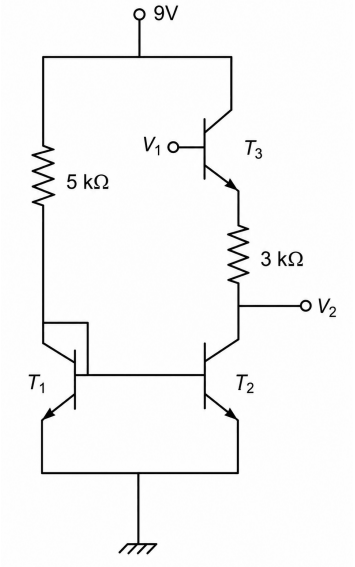


Fig. Q4.14.1

Q4.14.2 NAT 2014 1M

For the circuit shown in the figure, the transistor has $\beta = 40$, $V_{BE} = 0.7$ V, and the voltage across the Zener diode is 15 V.

The current (in mA) through the Zener diode is _____.

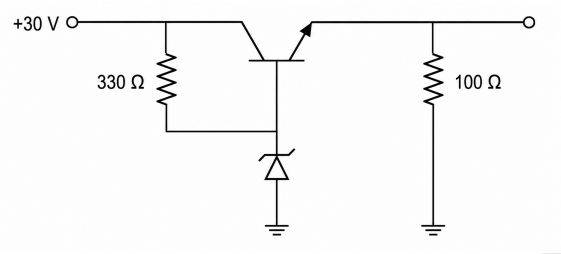


Fig. Q4.14.2

Q4.12.1 MCQ 2012 2M

The voltage gain A_v of the circuit shown below is

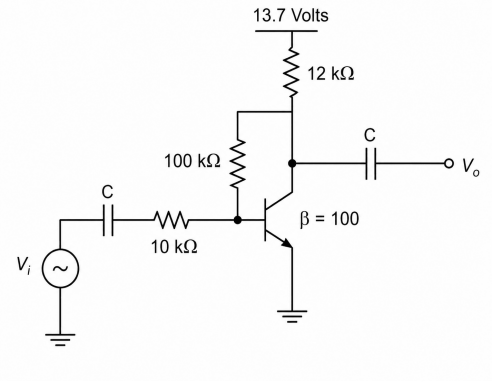


Fig. Q4.12.1

A. $|A_v| \approx 200$ **B.** $|A_v| \approx 100$
C. $|A_v| \approx 20$ **D.** $|A_v| \approx 10$

Q4.10.1 MCQ 2010 2M

The matched transistors Q_1 and Q_2 shown in the adjoining figure have $\beta = 100$. Assuming the base-emitter voltages to be 0.7 V, the collector-emitter voltage V_2 of the transistor Q_2 is

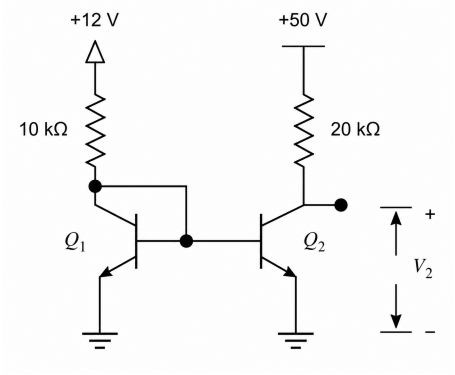


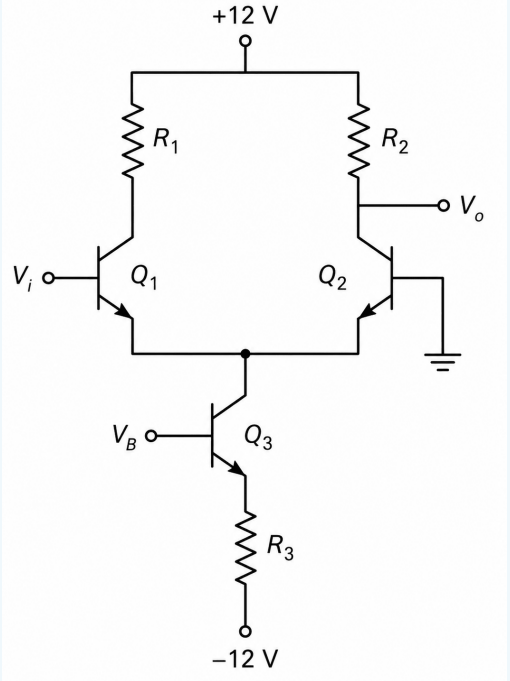
Fig. Q4.10.1

A. 33.9 V **B.** 27.8 V **C.** 16.2 V **D.** 0.7 V

Common Data Questions

The next 2 questions share the data below.

The circuit shown in the figure uses three identical transistors with $V_{BE} = 0.7$ V and $\beta = 100$.
Given $R_1 = R_2 = R_3 = 1$ kΩ, $kT/q_e = 25$ mV. The collector current of transistor Q_3 is 2 mA.



Q4.09.1 MCQ 2009 2M

The bias voltage V_B at the base of the transistor Q_3 is approximately

A. −9.3 V **B.** −10.0 V **C.** −10.3 V **D.** −11.0 V

Q4.09.2 MCQ 2009 2M

The small-signal voltage gain of the circuit is

A. −20 **B.** −40 **C.** 20 **D.** 40

Q4.08.1 MCQ 2008 2M

In the amplifier circuit shown below, assume $V_{BE} = 0.7$ V and the β of the transistor and the values of C_1 and C_2 are extremely high. If the amplifier is designed such that at the quiescent point its $V_{CE} = V_{CC}/2$, where V_{CC} is the power supply voltage, its small signal voltage gain $\left|\frac{V_{out}}{V_{in}}\right|$ will be

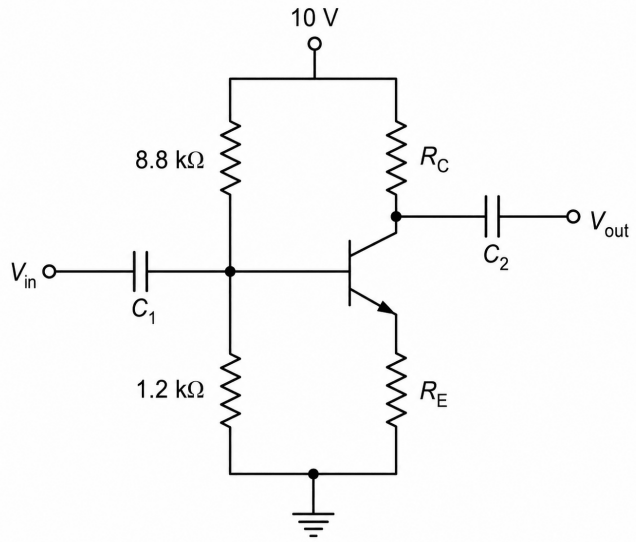


Fig. Q4.08.1

A. 3.75 **B.** 4.5 **C.** 9 **D.** 19

Q4.08.2 MCQ 2008 1M

For a single stage BJT common base amplifier,

A. current gain as well as voltage gain can be greater than unity

B. current gain can be greater than unity but voltage gain is always less than unity

C. voltage gain can be greater than unity but current gain is always less than unity

D. current gain as well as voltage gain is always less than unity

Q4.07.1 MCQ 2007 2M

The three transistors in the circuit shown below are identical, with $V_{BE} = 0.7$ V and $\beta = 100$.
The voltage V_c is

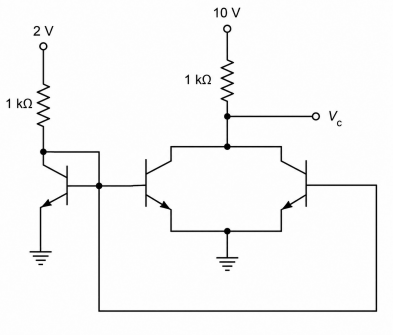


Fig. Q4.07.1

A. 0.2 V **B.** 2 V **C.** 7.4 V **D.** 10 V

Q4.07.2 MCQ 2007 2M

In the circuit shown below, $V_{BE} = 0.7$ V. The β of the transistor and V_{CE} are, respectively

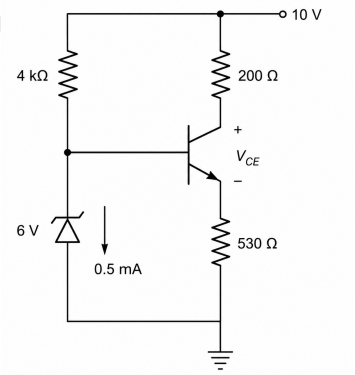


Fig. Q4.07.2

A. 19 and 2.8 V **B.** 19 and 4.7 V
C. 38 and 2.8 V **D.** 38 and 4.7 V

Q4.06.1 MCQ 2006 2M

An amplifier circuit is shown below. Assume that the transistor works in active region. The low frequency small-signal parameters for the transistor are $g_m = 20$ mS, $\beta_0 = 50$, $r_o = \infty$, $r_b = 0$.

What is the voltage gain, $A_v = \left(\frac{v_o}{v_i}\right)$, of the amplifier?

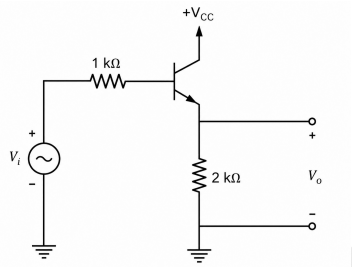


Fig. Q4.06.1

A. 0.967 **B.** 0.976 **C.** 0.983 **D.** 0.998

Q4.06.2 MCQ 2006 2M

The biasing circuit of a silicon transistor is shown below. If $\beta = 80$, then what is V_{CE} for the transistor?

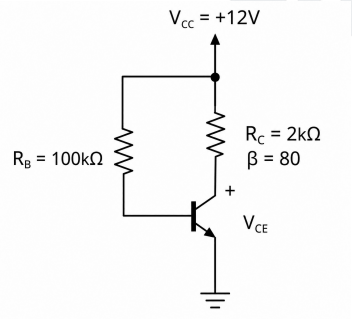


Fig. Q4.06.2

A. −6.08 V **B.** 0.2 V **C.** 1.2 V **D.** 6.08 V

Q4.03.1 MCQ 2003 2M

A transistor amplifier circuit is shown in figure. The quiescent collector current, rounded off to first decimal, is

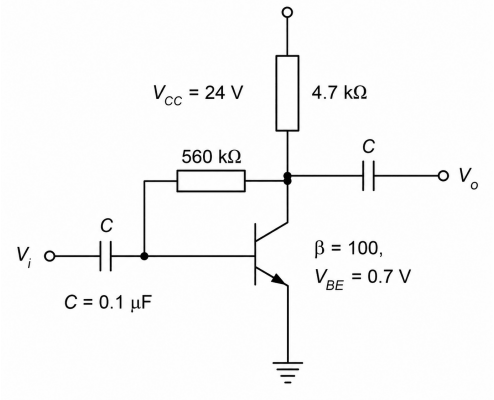


Fig. Q4.03.1

A. 2.6 mA **B.** 2.3 mA **C.** 2.1 mA **D.** 2.0 mA

Q4.00.1 MCQ 2000 1M

If both the junctions of a transistor are forward biased it will be in

A. saturation mode **B.** active mode
C. cut-off mode **D.** inverse active mode

Q4.99.1 MCQ 1999 2M

A unity gain buffer amplifier has a bandwidth of 1 MHz. The output voltage of the amplifier for an input of 2 V sinusoid of frequency 1 MHz will be

A. 2 V **B.** $2\sqrt{2}$ V **C.** $\frac{2}{\sqrt{2}}$ V **D.** $\frac{1}{\sqrt{2}}$ V

Q4.98.1 MCQ 1998 1M

For a silicon BJT shown in figure find R_B to establish $V_{CE} = 2$ V. Assume $V_{BE} = 0.7$ V.

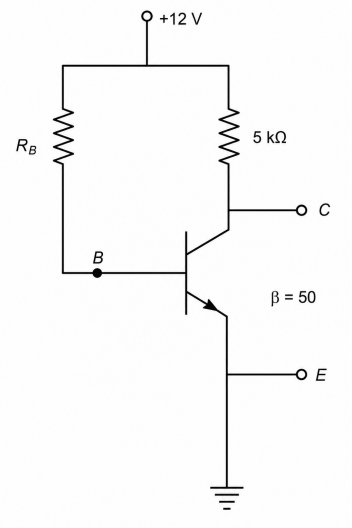


Fig. Q4.98.1

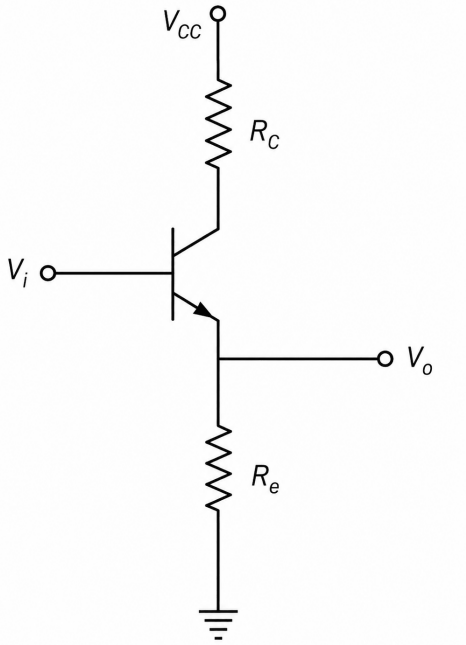


Fig. Q4.95.1

A. 283 kΩ **B.** 107 kΩ **C.** 200 kΩ **D.** 242 kΩ

Q4.97.1 MCQ 1997 1M

A common emitter transistor amplifier using a collector load of 1 kΩ and operating at room temperature with I_C = 1 mA, gives a voltage gain of nearly equal to

A. 25 **B.** 40 **C.** 250 **D.** 1000

Q4.95.1 Descriptive 1995 1M

The circuit shown in figure is said to be in common _____ configuration.

Q4.94.1 MCQ 1994 1M

The common collector transistor configuration has the following property:

A. High input and low output resistances

B. High input and high output resistances

C. Low input and low output resistances

D. low input and high output resistances

Q4.93.1 NAT 1993 1M

The input impedance of the emitter follower of figure shown is _____ $k\Omega$. h_{ie} = 600 Ω, h_{fe} = 89.

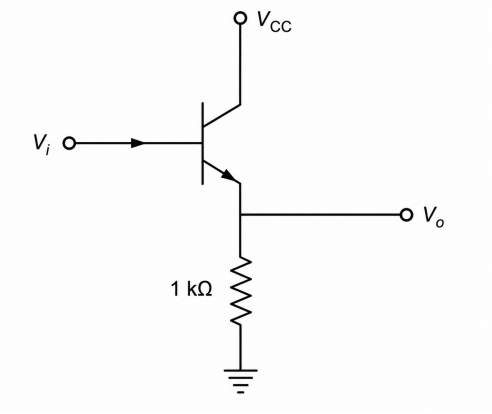


Fig. Q4.93.1

Q4.26.1 A	Q4.25.1 3.9 to 4.10	Q4.22.1 148.00 to 152.00
Q4.21.1 10.00	Q4.21.2 1	Q4.18.1 300
Q4.15.1 93	Q4.14.1 5.5 to 5.8	Q4.14.2 41 to 42
Q4.12.1 D	Q4.10.1 B	Q4.09.1 A
Q4.09.2 C	Q4.08.1 C	Q4.08.2 C
Q4.07.1 C	Q4.07.2 A	Q4.06.1 A
Q4.06.2 B	Q4.03.1 B	Q4.00.1 A
Q4.99.1 C	Q4.98.1 A	Q4.97.1 B
Q4.95.1 Collector	Q4.94.1 A	Q4.93.1 90.6
Q4.92.1 *		

Q4.92.1 NAT 1992 1M

The figure shown below is a common emitter amplifier. The component _____ is wrongly placed because _____.

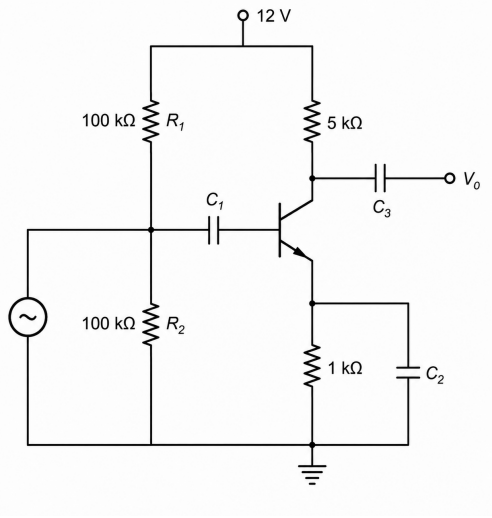


Fig. Q4.92.1

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

UNIT

V

Miscellaneous Topics

Topics

1. CMOS Inverter

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 4	**MCQ:** 3	**MSQ:** 1	**NAT:** 0
1M: 2	**2M:** 2	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2023)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MSQ 2026 2M

Which of the following functions represent(s) the output of the CMOS logic-gate shown in the circuit below?

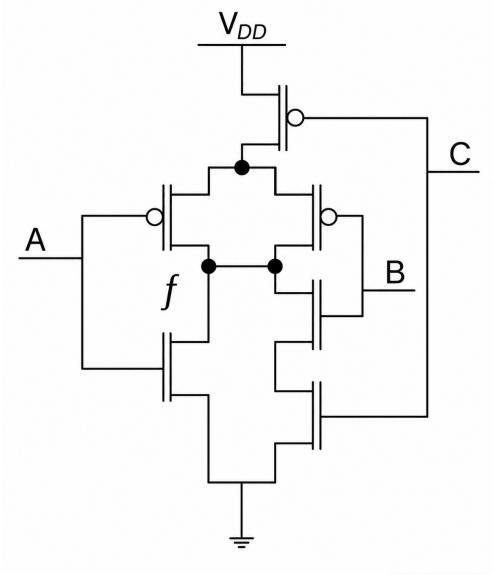


Fig. Q5.26.1

- **A.** $f = \overline{A + B \cdot \overline{C}}$
- **B.** $f = \overline{A + BC}$
- **C.** $f = \overline{A} \cdot \left(\overline{B} + \overline{C}\right)$
- **D.** $f = \overline{A} \cdot (B + C)$

Q5.23.1 MCQ 2023 1M

Choose the fastest logic family among the following:

A. Transistor-Transistor Logic

B. Emitter-Coupled Logic

C. CMOS Logic

D. Resistor-Transistor Logic

Q5.19.1 MCQ 2019 2M

A Voltage amplifier is constructed using enhancement mode MOSFETs labeled M1, M2, M3 and M4 in the figure below. M1, M2 and M4 are n-channel MOSFETs and M3 is a p-channel MOSFET. All MOSFETs operate in saturation mode and channel length modulation can be ignored. The low frequency, small signal input and output voltages are V_{in} and V_{out} respectively and the dc power supply voltage is V_{DD}. All n-channel MOSFETs have identical transconductance g_{mn} while the p-channel MOSFET has transconductance g_{mp}. The expressions for the low frequency small signal voltage gain $V_{\text{out}}/V_{\text{in}}$ is

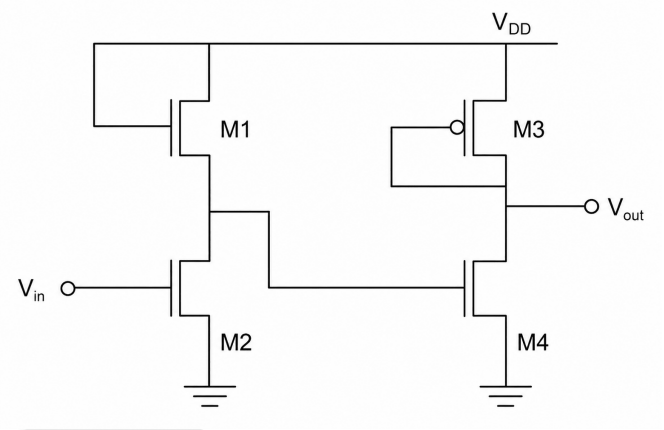


Fig. Q5.19.1

A. $-\dfrac{g_{mn}}{g_{mp}}$ **B.** $-g_{mn}\left(g_{mn}+g_{mp}\right)^{-1}$

C. $+\dfrac{g_{mn}}{g_{mp}}$ **D.** $g_{mn}\left(g_{mn}+g_{mp}\right)^{-1}$

Q5.14.1 MCQ 2014 1M

The figure is a logic circuit with inputs A and B and output Y. $V_{ss} = 5\text{V}$. The circuits is of type

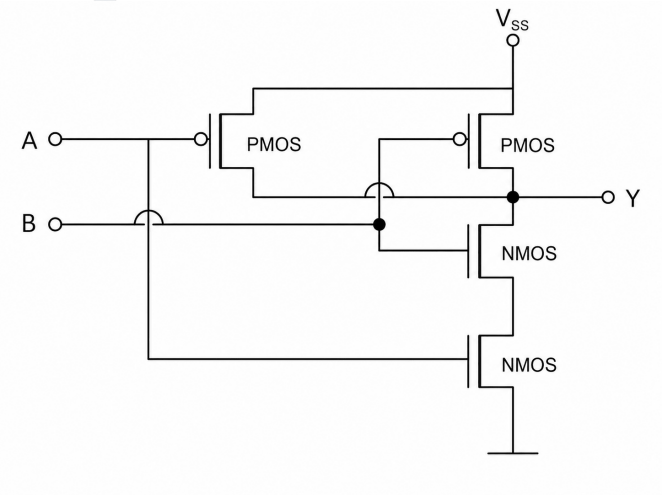


Fig. Q5.14.1

A. NOR **B.** AND **C.** OR **D.** NAND

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Q5.26.1 **MTA**	Q5.23.1 **B**	Q5.19.1 **C**
Q5.14.1 **D**		

PrepFusion

Also in this Series

Every PrepFusion GATE title is built the same way: the real previous-year papers, nothing paraphrased or reconstructed, arranged so the pattern the exam repeats is visible on the page. Each questions volume has a companion solutions volume that reuses the same question identifiers, so you can move between the two without a lookup table.

NexusX	Network Theory, Analog Electronics, Digital Electronics, Control Systems and Signals & Systems — EC, EE, IN
SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
General Aptitude	Verbal and quantitative aptitude, arranged year-wise and session-wise — all branches

Spotted a wrong answer key, a typo or a question we have mis-tagged? Report it on the PYQ portal at `pyq.prepfusion.in` — corrections go into the next printing, and you can request a video solution for any question from the same place.

PrepFusion